Supplementary Information

Joint phasing of sequence variants and DNA methylation from long reads with LongPhase 2

Jyun-Hong Lin, Yang-Ming Yeh, Yu-Chun Chen, Yao-Ting Huang

Contents

Supplementary Methods **2**

Supplementary Method 1. Allele-specific methylation calling (`modcall`) 2

Supplementary Method 2. Read-level allele observations and copy-number interval detection 2

Supplementary Method 3. Tuning of the voting and read-filtering parameters 4

Supplementary Method 4. Graph neural network: features, architecture, inference, training, threshold selection and export 4

Supplementary Method 5. Read haplotagging 7

Supplementary Method 6. Evaluation metrics 9

Supplementary Tables **10**

Supplementary Figures **30**

Supplementary Methods

Supplementary Method 1. Allele-specific methylation calling (`modcall`)

Modification interpretation. Only the 5mC probability (modification code `m`) was used. 5hmC calls (code `h`) were ignored; under a 5mCG_5hmCG model, a cytosine called as 5hmC was counted as unmethylated whenever its 5mC probability was ≤ 0.2. Reverse-strand calls were mapped to the forward-strand CpG coordinate, and the two strands were genotyped jointly. Non-CpG modification calls were discarded.

Linking modes and the depth bound on the neighbour scan. In *SNV-anchored* mode, used whenever an SNV VCF was supplied, a candidate CpG was accepted as a strong anchor if it satisfied the co-segregation criterion of Methods against an SNV among the next 20 variants (`-connectAdjacent`). In *methylation-only* mode, used where no informative SNV was in range, the same criterion was applied between two candidate CpGs. In both modes a pair was considered only while the number of reads spanning it exceeded $\max(6, (n_1 + n_2)/4)$, where n_1 and n_2 are the numbers of reads recorded at the two sites. The neighbour scan for a site stopped at the first pair at or below this bound. Candidates that never satisfied the co-segregation criterion were retained as weak candidates. They were admitted if they satisfied the criterion on at least six reads against an already accepted CpG, in two expansion rounds (`-iterCount`, default 2).

Merging runs of CpGs. Runs of consecutive CpG coordinates were merged into a single locus represented by the first position, and the strand-specific observations of the run were consolidated into one read list. These methylation loci were phased jointly with sequence variants. The resulting VCF records the methylated and unmethylated read names (`MR`, `NR`), the read strand and the per-site depths, and is passed to `phase` through `-mod-file`.

Supplementary Method 2. Read-level allele observations and copy-number interval detection

Variant eligibility and repeat annotation. Homozygous and filtered small-variant records were carried through to the output but did not enter the phasing graph. An optional indel-quality filter excluded indels below a user-specified QUAL threshold and marked their FILTER field; the default threshold of 0 disabled this filter. An indel was flagged as a tandem-repeat indel when the dinucleotide immediately downstream of its position occurred in five consecutive copies. Flagged indels remained in the graph and cast votes at one-tenth weight. Homozygous SVs, SVs whose breakpoint coincided with an SNV and SVs with duplicated positions were excluded. The supporting read names and length of each retained SV were recorded.

Per-class allele extraction. SNV alleles were read from the aligned base and inherited its Phred base quality. Insertion and deletion alleles were called from the CIGAR string: a read supports the ALT allele if a CIGAR insertion or deletion begins immediately after the variant position, and the REF allele otherwise; when a read deletion spans an SNV in a homopolymer of length ≥ 3, the SNV allele was recovered from the base following the deletion. SV alleles were assigned by scanning a window of 20 CIGAR operations (`-svWindow`) around the SV start for an insertion or deletion whose length is within 10% of the SV length (`-svThreshold`); reads listed by the SV caller as supporting the SV were also assigned the ALT allele. 5mC alleles were taken from the `modcall` read lists (methylated = allele 0, unmethylated = allele 1) restricted to reads whose strand matched the recorded strand. Soft- and hard-clipping operations longer than 5 bp were counted per reference position, separately for clips at the front (left) and back (right) end of an alignment, as evidence of alignment breakpoints for copy-number detection (below).

The copy-number-aware filter (Methods; Supplementary Fig. 1e,f) requires the boundaries of segments that recruit reads from more than one locus. These boundaries were detected from clusters

of soft- and hard-clips by a scan of the per-position clipping counts of each chromosome in coordinate order.

Input. For every reference position at which at least one alignment carries a clipping operation longer than 5 bp, two counts are recorded: u, the number of alignments clipped at their front (left) end, and d, the number clipped at their back (right) end. An interval still open at the end of the chromosome was evaluated at a position 30 kb beyond the last observed clip.

State. The scanner is in one of three states: *idle*, *step* (a copy-number transition, entered on a strong front-clip cluster) and *ramp* (progressively nested breakpoints, entered on a weak front-clip excess). In the two active states it carries a running excess c of front- over back-clips, a candidate start, a candidate end and three adaptive thresholds derived from the front-clip count n that opened or re-anchored the interval:

$$r = n, \qquad p = \begin{cases} \lfloor n/2 \rfloor, & n \geq 10 \\ 5, & n < 10 \end{cases},$$

$$f = \begin{cases} 5, & n \geq 20 \\ \lfloor n/4 \rfloor, & 10 \leq n < 20 \\ 2, & n < 10, \end{cases}$$

where r is the re-anchor threshold, p the pull-down threshold and f the decay floor.

Transitions. With constants $\Pi = 5$ (push threshold), $A = 30{,}000$ bp (look-ahead), $E = 30$ (extend threshold), $\Lambda = 200{,}000$ bp (maximum interval length) and $\Sigma = 20$ (ramp commit threshold), the scanner at position x with counts (u, d) behaves as follows:

```
idle:
  if u >= PI and c == 0:
      enter STEP;  c = u - d;  start = x;  end = x + A
      set r, p, f from u
  elif u > d and c == 0:
      enter RAMP;  c = u - d;  start = x;  end = x + A

step:
  if u > r:                       # stronger breakpoint: re-anchor
      start = x;  end = x + A;  set r, p, f from u
  c += u - d
  if c > E:  end = x + A
  if d >= p:                      # pull-down reached
      emit [start, x];  reset to idle
  elif c <= f and x <= end:       # decayed to the floor in range
      emit [start, x];  reset to idle
  if x > end or c <= 0 or x - start >= LAMBDA:
      reset to idle               # candidate discarded

ramp:
  if (c > SIGMA and d >= c/4) or (c <= SIGMA and d >= PI):
      emit [start, x];  reset to idle
  elif u >= PI:                   # promote to STEP
      c = u - d;  start = x;  end = x + A;  set r, p, f from u
  else:
      c += u - d
      if c > E:  end = x + A
      if x > end or c <= 0 or x - start >= LAMBDA:
          reset to idle           # candidate discarded
```

A stronger front-clip cluster re-anchored a *step* interval: the candidate start moved to the new position and the three thresholds were recomputed. The reported interval began at the strongest breakpoint. The decay-floor exit applied only within the look-ahead window ($x \leq$ end); candidates outside this window were discarded.

Site removal. Within each emitted interval the mismatch-rate statistic $\mathrm{MR}(v)$ defined in Methods is computed and heterozygous variants with $\mathrm{MR}(v) \geq 0.7$ are removed from all reads. A variant is scored only when both allele populations have a non-zero mean load; a site at which one population carries no other alternate allele is not removed on this basis.

Rationale for the mismatch-load statistic. Comparable alternate-allele loads in the two read groups give MR near 0.5. Larger loads in the reads that carry the alternate allele increase MR and may indicate reads misaligned from paralogous sequence.

Choice of constants. The constants of the scanner (Π, A, E, Λ and Σ, those of the adaptive thresholds and the 5-bp minimum clip length) were set by inspection of clipping profiles in copy-number-variable segments of HG002, the individual evaluated here, in which many false heterozygous SNVs had been called; they were not tuned by a parameter sweep. The look-ahead window A approximates the read length (read N50 of 29 kb; Methods). The thresholds are absolute alignment counts applied unchanged at every coverage, and the sensitivity of the results to them was not assessed.

Supplementary Method 3. Tuning of the voting and read-filtering parameters

The voting and read-filtering constants were set empirically: the near-certain and exclusion ambiguity thresholds of 0.1 and 0.7, respectively, the near-certain vote weight of 20, the base-quality threshold of 12, and the single-read guard thresholds of 3 votes and 0.2 ambiguity. The voting rule and single-read guard are described in Methods, "Haplotype resolution by multi-neighbour voting"; base-quality weighting is described in "Weighted phasing graph".

Each constant was fixed by a sweep on HG002 at 10–60× before the experiments reported here, and none was re-tuned afterwards. The sweeps were run during the development of LongPhase in 2023–2024 on whole-genome phasing of earlier HG002 nanopore alignments, before the alignments evaluated here were generated and before the chromosome split used for GNN training was defined. The parameters were therefore tuned on data from the same individual as the evaluation, and chr17, chr21 and chr22 are held out from GNN training but not from this tuning.

Supplementary Method 4. Graph neural network: features, architecture, inference, training, threshold selection and export

The GNN predicts label-defined phase-disagreement onsets, comprising isolated disagreements and the first positions of consecutive disagreement runs. Its error-class output is used as an onset score for selective unphasing.

Node features. Each allele node carried 31 features in four groups, phasing evidence, genomic context, graph structure and variant class, defined in Supplementary Table 1 (window construction in Supplementary Figs. 4 and 5). SNVs and indels were distinguished by $||\mathrm{ALT}| - |\mathrm{REF}|| > 0$.

Edge features. Edges were taken from the DOT graph, restricted to pairs both of whose endpoints lie inside the window, symmetrized by copying the features of whichever direction was present, and augmented with a self-loop per node whose features are the mean of its non-self incident edges. The vote weight w of an edge is the weighted haplotype vote recorded for the source allele when the vote was cast. Each edge carried six features (Supplementary Table 2).

Architecture. The network used the GPS graph-transformer architecture[1] (Supplementary Table 3 and Supplementary Fig. 6). Node features were standardized with a batch-normalization layer whose statistics are frozen at inference, linearly projected to a 128-dimensional embedding and passed through a GELU non-linearity. Four identical layers then each combine a local and a global branch. The local branch is a four-head GATv2 message-passing layer[2,3] in which the attention logit of head t for edge $j \to i$ is

$$\alpha_{ij}^{(t)} \;\propto\; \exp\Big(\mathbf{a}_t^{\top}\,\mathrm{LeakyReLU}_{0.2}\big([\mathbf{W}_r\mathbf{x}_i + \mathbf{W}_l\mathbf{x}_j + \mathbf{W}_e\mathbf{e}_{ij}]_t\big)\Big), \tag{1}$$

where $[\cdot]_t$ selects the 32 coordinates of head t, $\mathbf{W}_e$ is a per-layer edge encoder without bias and the normalization is a softmax over the incoming edges of i; the head outputs are concatenated and a bias is added. The global branch is four-head scaled dot-product self-attention over all nodes of the window[4], with no masking. The two branch outputs are added to the residual stream, followed by layer normalization, a two-layer feed-forward block (128→256→128, GELU) with a second residual connection and layer normalization. Two edge-context features were concatenated with the final node embedding: the sum and the non-negative maximum of the logistic-transformed vote weights over incoming edges, excluding the self-loop. For an edge with vote weight w, the transformed weight is $1/(1+e^{-w})$. The pooled vector is then passed through a two-layer classifier (130→128→2) with softmax output. The model comprises 687,358 stored values, including the frozen batch-normalization statistics.

Scoring zone and inference. Inference was implemented in C++ without external runtime dependencies and used the erf-based GELU. Predictions for the two allele nodes of a variant were averaged. Only variants in the central zone of each window were scored: those whose genomic distance from the trigger is at most half the largest such distance in the window, $|p - p_c| \le \frac{1}{2}\max_u |p_u - p_c|$. Only labelled variants in this zone contributed to the training loss, and the same zone was scored at inference; the other variants supplied graph context. A variant scored in several windows received the running mean of its predictions from those windows.

Phase-set update. A variant with a mean onset score ≥ 0.35 was unphased while preserving its alleles: both `0|1` and `1|0` became `0/1`, and the PS tag was removed. Bridge protection was disabled in the reported runs. Connectivity was then evaluated over the surviving graph edges within each original phase set. The largest connected component retained the original PS identifier; each other component received the 1-based position of its first variant as its identifier. Corrected records were written separately for small variants, SVs and methylation sites.

Labels. Labels were derived from the v5.0q benchmark restricted to chr1–chr22 small variants, the same truth set used for evaluation, and describe the phase of a variant relative to its neighbours rather than in absolute terms. Within each window the variants were grouped by phase set; in each group the first variant present in the benchmark fixed the orientation of that group and was labelled *correct*, and every later variant was labelled *correct* if its allele-to-haplotype orientation agreed with that anchor and *disagreeing* otherwise. Each maximal run of consecutive disagreeing variants was attributed to its initiating event rather than counted as a separate error at each variant. A run of length one became *unphase* if that variant's PE was at least 0.9 and *flip* otherwise, whereas a run of length two or more had its first variant, the switch point, relabelled *break*, and all remaining members reset to *correct*. Variants absent from the benchmark, and all SV and 5mC records, for which the benchmark carries no phase truth, were left unlabelled: they remained in the graph as context but did not contribute to the loss. The released model was trained and applied in binary mode. The binary classifier merged *flip*, *unphase* and *break* into a single *error* class; predicted targets were subsequently unphased.

Class balance and loss. Among the labelled central nodes of the training split, 17,812,710 (97.04%) were *correct* and 544,170 (2.96%) *error*, and 21.5% of windows contained at least one

error node; the prevalence is far above the genome-wide switch-error rate because windows are opened only at high-entropy triggers and because the label is assigned relative to a window-local anchor rather than to a whole phase block. Each minibatch kept all of its error nodes and a uniform random sample of correct nodes of at most four times that number (`-ratio_corr 4`), drawn again for every minibatch. The cross-entropy on that subset was focal-weighted [5] by $(1-p_t)^\gamma$ with $\gamma = 1.5$ and used label smoothing of 0.05.

Training augmentations. Two label-preserving augmentations were applied to training windows only, each independently with probability 0.5. Haplotype swapping exchanged the two allele nodes of every variant and complemented the HP feature. After the exchange, each edge received the vote weight w of its new source node. Coverage down-scaling multiplied h_1, h_2, their sum, the outgoing-weight standard deviation and the raw edge weights by a factor drawn uniformly from $[0.4, 0.7]$, then recomputed the derived logarithmic features; it was applied only to windows containing at least one error node. In release v2.1, swapped windows retained the pre-exchange source-node weights on their edges, so the edge weights did not match their source nodes. The model was retrained with the corrected augmentation for v2.1.1.

Model selection. Checkpoints were selected on the validation chromosomes by a normalized net-gain criterion. At each epoch the predicted onset score was thresholded at each of 81 values spanning $[0.01, 0.99]$. For $P > 0.5$, the score was

$$S = R\left(2 - \frac{1}{P}\right) = \frac{\mathrm{TP} - \mathrm{FP}}{\mathrm{TP} + \mathrm{FN}}, \tag{2}$$

where $P = \mathrm{TP}/(\mathrm{TP} + \mathrm{FP})$ and $R = \mathrm{TP}/(\mathrm{TP} + \mathrm{FN})$ are precision and recall for the label-defined error class. For a fixed evaluation set, $\mathrm{TP} + \mathrm{FN}$ is constant. Thus, maximizing S is equivalent to maximizing $\mathrm{TP} - \mathrm{FP}$, assigning unit benefit to each true-positive detection and unit cost to each false-positive prediction. This score serves as a proxy for correction utility, rather than an exact fraction of switch errors removed from the corrected VCF. The score was maximized over thresholds subject to recall of at least 0.005; for $P \leq 0.5$, the score was instead $P - 1$. The selected checkpoint was epoch 30 ($S = 0.0221$ at $P = 0.621$, $R = 0.057$ and a threshold of 0.574); its classification metrics on the validation and held-out chromosomes are given in Supplementary Table 4. The score S governed checkpoint selection; the deployment threshold was chosen separately using the end-to-end phasing trade-off described below.

Label definitions, seeds and deployment threshold. Two label definitions were trained with the settings above, each with five seeds (42, 43, 44, 1337 and 3407): benchmark variants matched to the calls by position, as described under Labels, and matched by position, REF and ALT, which leaves calls whose alleles differ from the benchmark unlabelled. Every model was applied to 60 HG002 nanopore SNV-only runs, comprising ten down-sampling replicates at each of 10, 12, 14, 16, 18 and 20×, and scored relative to phasing without GNN correction by the fraction of switch errors removed and by the cost of correction, the number of phased SNVs withheld divided by the number of switch errors removed. Because the two trade off against each other through the onset-score threshold B, each model's threshold was set on the validation chromosomes chr15 and chr16 to the same cost, 6.35 withheld SNVs per removed error (the cost of the release v2.1 model at $B = 0.30$), and the models were compared at that cost (Supplementary Table 19). The REF/ALT-matched labels gave higher classification scores on the held-out chromosomes (error-class F_1 0.367–0.382 against 0.345–0.355) but removed fewer switch errors there (mean of five seeds 21.6% against 23.0%), and over the whole genome (33.5% against 33.8%). The position-matched labels were therefore kept. Among their five seeds, the released model is the one that removed the most switch errors on the validation chromosomes (seed 42, 50.8%); the threshold selected on the validation chromosomes was 0.34 and was rounded to $B = 0.35$ for deployment. With the released implementation and $B = 0.35$, correction removed 34.3% of switch errors at a cost of 7.98 withheld

SNVs per removed error (60 runs pooled, whole genome), and 23.5% at a cost of 9.26 on the held-out chromosomes chr17, chr21 and chr22 (Supplementary Table 15). Neither the label definition nor the threshold was selected on the held-out chromosomes. This criterion concerns the final phasing output and is distinct from the classifier-based criterion used to rank training checkpoints.

Export and numerical agreement. The trained model was rewritten with dense $N \times N$ adjacency and edge tensors, exported to ONNX (opset 17, dynamic N) and compiled into the C++ module; the logistic transform of the vote weight and the self-loop features are applied to the inputs before inference. Agreement was checked at each step: the dense rewrite against the PyTorch Geometric model and the ONNX graph against the dense rewrite, on a real training window, with an acceptance bound of 10^{-3} on the absolute difference in onset score; and the compiled C++ module against ONNX Runtime on six synthetic graphs of 3–120 nodes at edge densities 0.1–0.4, where the largest absolute difference was 3×10^{-7} (acceptance bound 10^{-5}).

Supplementary Method 5. Read haplotagging

`longphase haplotag` assigns each alignment to a haplotype using the phased VCFs produced by phasing with or without GNN correction.

Alignment selection. Primary alignments with mapping quality ≥ 1 (`-q, -qualityThreshold`) are considered. Supplementary alignments are tagged only under `-tagSupplementary`; secondary alignments never are. Any pre-existing `HP`, `PS` and `PQ` tags are removed from every record before tagging. Unmapped records are passed through unchanged.

Evidence collection. Each alignment is traversed once along its CIGAR, and four classes of evidence are collected into two per-haplotype tallies (n_1, n_2):

- *SNVs* are read from `M`, `=` and `X` operations; a read carrying the allele phased to haplotype k adds 1 to n_k.
- *Small indels* are scored by allele length rather than by sequence comparison. Writing ℓ_1 and ℓ_2 for the lengths of the two phased alleles at the site, a read that carries an indel there supports the haplotype whose allele has $\ell \neq 1$, and a read that does not supports the haplotype whose allele has $\ell = 1$. When both or neither allele has length 1 the site casts no vote.
- *SVs* are re-derived from the alignment, not taken from the caller's read list: a window of $\pm$`-svWindow` CIGAR operations (default 20) around the current operation is scanned for an insertion or deletion whose length is within `-svThreshold` (default 0.10) of the SV length, and the first match adds 1 to the haplotype the SV is phased to.
- *Allele-specific 5mC* sites are matched by read name, not by position: the methylated-read list `INFO/MR` of the phased 5mC VCF adds 1 to the haplotype that record is phased to, for each read it names. The unmethylated list `NR` is not consulted; a 5mC site contributes evidence only to the reads that are methylated at it.

Each informative observation contributed unit weight during haplotagging, whatever its class. During read-based correction inside `phase`, by contrast, SNV and SV alleles contribute 1, indel and tandem-repeat-indel alleles 0.1, and 5mC alleles are excluded (Methods).

Decision and phasing quality. Let $m = \max(n_1, n_2)$ and $\mu = \min(n_1, n_2)$. The read is assigned to the majority haplotype when $m/(m+\mu) \geq$ `-p, -percentageThreshold` (default 0.6) and is otherwise left untagged. A Phred-scaled phasing quality is computed for every alignment as

$$\mathrm{PQ} = \begin{cases} 0 & m = 0, \\ 40 & \mu = 0, \\ \left\lfloor -10 \log_{10} \dfrac{\mu}{m+\mu} \right\rfloor & \text{otherwise}, \end{cases} \tag{3}$$

`PQ` is written as an integer auxiliary tag `PQ:i` alongside `HP:i` and `PS:i`, and only on tagged reads.

Phase-set veto. The phase sets of all informative variants on the read are counted. Reads with informative variants in more than one phase set were left untagged, regardless of their haplotype tally. The `PS` value written on a tagged read is that of the first phased variant at or after the alignment start, whether or not that variant is informative on the read. The output `PS` normally matches the phase set of the informative variants because the veto requires all informative variants to share one phase set. It can differ when the first phased variant at or after the alignment start is uninformative on the read and belongs to another block.

Supplementary Method 6. Evaluation metrics

Metrics were computed with `longphase compare` (Supplementary Fig. 8), a C++ re-implementation of the `whatshap compare` algorithm[6].

Block intersection. For each chromosome, heterozygous records common to the truth and a query are intersected by position and by REF/ALT alleles. The intersection is partitioned into maximal runs that lie in one truth phase set and one query phase set simultaneously; runs of length one carry no phase information and are discarded. Switch errors and Hamming distance are computed within the surviving blocks.

Switch errors and flips. Each block yields two binary strings, the truth haplotype assignment p_0 and the query assignment q_0 of its variants. Both are converted to *switch encoding*, the string of relative changes between consecutive positions, which removes the arbitrary orientation of the block. Switch errors are the Hamming distance between the two switch-encoded strings. The decomposition into isolated switches and flips follows WhatsHap's `compute_switch_flips`: a maximal run of k consecutive disagreements in switch space is charged as $\lfloor k/2 \rfloor$ flips and $k \bmod 2$ switches. A single variant assigned to the wrong haplotype thus counts as one flip, not two switches. The reported SNV *switch error rate* is the total number of SNV switch errors divided by the number of assessed consecutive SNV pairs.

Hamming distance. For each block the query strings (q_0, q_1) are compared with the truth strings (t_0, t_1) in both orientations, and the smaller of $\mathrm{H}(t_0, q_0) + \mathrm{H}(t_1, q_1)$ and $\mathrm{H}(t_0, q_1) + \mathrm{H}(t_1, q_0)$ is halved to give the number of misassigned variants in that block. Block values are summed and expressed as a percentage of all compared variants. This denominator includes every variant class present in both files, including indels when present, unlike the SNV-pair denominator of the reported switch error rate.

Phased fraction. Phased fraction is counted before partitioning the truth–query intersection into blocks. Its numerator is the number of truth heterozygous variants whose position, REF and first ALT match a query record that is heterozygous and phased. Its denominator is all truth heterozygous variants of the same class.

Block N50. Query phase sets with fewer than two variants are excluded. The span of each remaining block is the distance in bp between its first and last variants. Block spans are sorted in decreasing order, and N50 is the span at which the cumulative sum first reaches half the total, which is the span-weighted median. It is computed once per chromosome and once over the block spans of all chromosomes pooled. The values plotted in Figs. 2c,f,i,l are the pooled, genome-wide N50 of the query. This metric uses query phase sets independently of the truth set and does not require truth–query intersection.

Supplementary Tables

Supplementary Table 1: Node features of the graph neural network. Each allele node of a window carries 31 features; # is the 1-based index in the feature vector, listed here by group, not by index. h_1, h_2: weighted haplotype votes (taken from the DOT edge labels when available, otherwise from `INFO/H1`, `INFO/H2`); PE: phasing entropy; PS: phase set; "centre" is the trigger variant of the window. Genomic-context features are derived from the reference sequence and input variant positions.

#	Feature	Definition
Phasing evidence (16)		
1	PE	normalized Shannon entropy of (h_1, h_2), 0–1 bit
2	HP	haplotype of this allele node (genotype index)
3	gt_ref	first component of the phased input genotype (GT)
4	gt_alt	second component of the phased input genotype (GT)
5	is_center	1 if the variant is the window trigger
6	is_same_ps	1 if the variant shares the trigger's PS
7	allele	0 for the reference-allele node, 1 for the alternate-allele node
8	rel_offset	$(\text{pos} - \text{pos}_c)$ / $\max\lvert\text{pos} - \text{pos}_c\rvert$ over the window, signed
9	H1	h_1
10	H2	h_2
11	total_votes	$h_1 + h_2$
12	vote_ratio	$\max(h_1, h_2)/(h_1 + h_2)$
13	log_total	$\log_2(1 + h_1 + h_2)$
14	log_h1	$\ln(1 + h_1)$
15	log_h2	$\ln(1 + h_2)$
25	rel_vote_depth	$\log_2\left((h_1 + h_2)/\overline{(h_1 + h_2)}_{\pm 10\,\text{variants}}\right)$
Genomic context (6)		
16	hp_len	homopolymer run length at the variant (capped at 20)
17	gc_content	GC fraction in ±50 bp
18	str_context	longest tandem-repeat unit count (unit 2–4 bp) in ±15 bp, /10, capped at 1
19	seq_entropy	Shannon entropy of the ±15 bp sequence (bits)
20	neigh_dens	(number of variants within ±500 bp − 1)/20
24	dist_near_indel	$\log_{10}$(1+ distance to nearest indel), capped at $\log_{10}(10001)$
Graph structure (5)		
21	out_degree	number of distinct target positions of outgoing DOT edges, /10, capped at 1
22	ps_block_size	ln(1+ variants in the PS)/ ln(101)
23	log_center_dist	$\log_{10}(1 + \lvert\text{pos} - \text{pos}_c\rvert)$
26	out_weight_std	s.d. of outgoing DOT edge weights, /100, capped at 1
27	is_bridge	1 if removing the variant disconnects the window graph
Variant class (4, one-hot)		
28–31	is_SNV, is_indel, is_SV, is_5mC	variant class indicators

Supplementary Table 2: Edge features of the graph neural network. Edges are taken from the DOT export, restricted to node pairs inside the window, symmetrized, and complemented by one self-loop per node whose features are the mean over the node's incident edges. w is the DOT edge label (accumulated vote weight).

#	Feature	Definition
1	weight	$1/(1+e^{-w})$
2	log_dist	$\log_{10}(1+\lvert \mathrm{pos}_{\mathrm{dst}} - \mathrm{pos}_{\mathrm{src}} \rvert)$
3	log_weight	$\ln(1+w)$
4	is_same_hp	1 if the two allele nodes have the same allele index
5	is_inter_ps	1 if the two variants belong to different phase sets
6	weight_ratio	w / (sum of outgoing edge weights of the source allele node)

Supplementary Table 3: Architecture and inference hyperparameters of the graph neural network.

Component	Setting
Input normalization	BatchNorm over 31 node features (frozen statistics at inference)
Input projection	Linear $31 \to 128$, GELU
Layers	4 GPS layers, hidden size 128
Local branch	GATv2, 4 heads $\times$ 32, LeakyReLU slope 0.2, per-layer edge encoder $6 \to 128$
Global branch	multi-head self-attention, 4 heads, Q/K/V/O projections $128 \to 128$
Fusion	$x \leftarrow x + \mathrm{local} + \mathrm{global}$; LayerNorm ($\varepsilon = 10^{-5}$)
Feed-forward	$128 \to 256 \to 128$, GELU, residual, LayerNorm
Classifier	concat$[h_i, \sum_j \sigma(w_{ij}), \max_j \sigma(w_{ij})]$ (130) $\to 128$ (GELU) $\to 2$, softmax; σ logistic, self-loop excluded
Stored values	687,358, including frozen normalization statistics
Window	trigger PE ≥ 0.8; ± 20 variants (≤ 41 variants, ≤ 82 allele nodes)
Decision	unphase if the mean onset score over the two allele nodes and windows in which the variant lies in the scoring zone is ≥ 0.35
Post-processing	split disconnected phase sets (largest component keeps PS); bridge protection off
Training	see Supplementary Table 4

Supplementary Table 4: Training configuration and evaluation of the released model. Values recorded in the released checkpoint and its training log. "Scored nodes" are labelled allele nodes inside the central zone of a window, the only nodes that contribute to the loss.

Item	Setting
Data	
Source runs	HG002 ONT R10.4.1, four-class co-phasing, 10/12/14/16/18/20× × 10 down-sampling replicates (60 runs)
Window extraction	`-pe_threshold 0.8`, `-window 20`, GRCh38 no-alt analysis set
Truth set	v5.0q benchmark, chr1–chr22 small variants
Split (by chromosome)	train chr1–chr14, chr18–chr20; validation chr15, chr16; test chr17, chr21, chr22
Windows / scored nodes	train 602,788 / 18,356,880; val 67,880 / 1,509,576; test 51,448 / 1,367,580
Class prevalence (train)	correct 17,812,710 (97.04%), error 544,170 (2.96%); 21.5% of windows contain an error node
Labels	anchor-relative within each PS; single disagreement → *unphase* if PE ≥ 0.9 else *flip*; run of ≥ 2 → first variant *break*, rest *correct*; merged to one *error* class
Masked from loss	variants absent from the benchmark; all SV and 5mC records (the truth set holds small variants only); nodes outside the central zone
Optimization	
Loss	focal cross-entropy, $\gamma = 1.5$, label smoothing 0.05, on a per-batch subset of all error nodes plus $\leq 4\times$ as many randomly drawn correct nodes
Optimizer	AdamW, $\beta = (0.9, 0.999)$, $\varepsilon = 10^{-8}$, weight decay 10^{-3}
Learning rate	3×10^{-4}, no warm-up, cosine annealing to 10^{-5} over 100 epochs
Batch size	64 windows, batched as disjoint graphs
Regularization	dropout 0.4 (local branch, global branch, FFN; inactive at inference); gradient-norm clipping 1.0
Graph preprocessing	reverse edge added per DOT edge with identical features (undirected)
Augmentation (train only)	haplotype swap with edge weights of the new source node, $p = 0.5$; coverage down-scaling $\times U[0.4, 0.7]$, $p = 0.5$, error-bearing windows only
Epochs	budget 100, early stopping after 25 epochs without improvement; stopped at 55
Seed	42 (Python and PyTorch generators); four further seeds and a second label definition in Supplementary Table 19
Environment	PyTorch 2.11.0, PyTorch Geometric 2.7.0, Python 3.12, CUDA 13.0, cuDNN 9.19, one NVIDIA GeForce RTX 5090
Selection and export	
Selection criterion	max over 81 thresholds in $[0.01, 0.99]$ of $R\,(2 - 1/P)$ subject to $R \geq 0.005$, on chr15–chr16; $P - 1$ where $P \leq 0.5$
Selected checkpoint	epoch 30; $S = 0.0221$ at $P = 0.621$, $R = 0.057$, threshold 0.574
Held-out test (chr17, 21, 22)	argmax: accuracy 0.936, error-class $P/R/F_1 = 0.251/0.564/0.347$; tuned threshold: 0.950, 0.294/0.471/0.362; $n = 41{,}378$ error nodes
Deployment threshold	0.35; calibrated to a validation cost of 6.35 withheld SNVs per removed switch error (Supplementary Method 4)
Export	ONNX opset 17, dynamic N, compiled into the C++ module; C++ vs. ONNX Runtime max. absolute difference 3×10^{-7}

Supplementary Table 5: Software and data used in the benchmarks.

Item	Version	Notes
HG002 nanopore R10.4.1 reads	ONT `giab_lsk114_2022.12`	FLO-PRO114M, SQK-LSK114, 4 kHz; Dorado `sup@v4.0.0` with `5mCG_5hmCG@v2`
Reference	GRCh38	`GCA_000001405.15_GRCh38_no_alt_analysis_set`; no-alt analysis set, no decoys
minimap2	2.24-r1122	`-y -ax map-ont` (alignment by Oxford Nanopore; reads with mean quality ≥ 10)
PEPPER-Margin-DeepVariant[7]	r0.8	SNV and indel calls (unphased genotypes); `call_variant` with `-ont_r10_q20` (nanopore) or `-hifi` (HiFi), one call set per coverage and replicate
Sniffles2	2.8.0	`-output-rnames`
GIAB v4.2.1 HG002	v4.2.1	GRCh38 small-variant benchmark with phase transferred from a hifiasm assembly; `HG002_GRCh38_1_22_v4.2.1_benchmark_hifiasm_v11_phasetransfer.vcf.gz`, biallelic phased heterozygous records
GIAB v5.0q HG002 benchmark[8,9]	v5.0q	GRCh38 small-variant benchmark derived from the T2T-HG002 v1.1 assembly; `HG002_GRCh38_v5.0q_smvar.vcf.gz`, chr1–chr22 records
WhatsHap	2.8	`phase -ignore-read-groups -reference`; `-only-snvs` for SNV comparisons; default indel phasing for indel comparisons
HapCUT2	1.3.4	`extractHAIRS -ont 1` (`-pacbio 1` for HiFi), then `HAPCUT2` with default parameters (default pruning); input restricted to heterozygous SNVs; with `extractHAIRS -indels 1` on the full call sets for the comparison with indels (Supplementary Table 18)
Margin	2.3.1	SNV-only comparison, 10–20×; phasing parameters `allParams.phase_vcf.ont.json`; input restricted to SNVs
Ralphi	v0.0.1 (commit `d7d17b1`)	SNV-only comparison, 10–20×; model `ralphi.v1.long.pt`, ONT configuration with `max_coverage 15`, `mapq 20`, read selection and bad-read filtering
MethPhaser	0.0.4	run on the uncorrected SNV-only LongPhase 2 output (same SNV phasing as the joint SNV+5mC run), replicate 1, with its alignments haplotagged by `longphase haplotag` (v2.0.2) and restricted to primary records; `meth_phaser_parallel` with that VCF (`-vc`) and a GTF of its phase blocks (`-g`), then `meth_phaser_post_processing`, otherwise default parameters; one local fix (when the block of a record cannot be found from its PS, which in LongPhase 2 output can precede the first phased variant of the block, it is found from the record position; 0.0.4 otherwise stops with an error), and output records split at missing line breaks and sorted
HapBridge[10]	commit `ec48f02`	nanopore, replicate 1, 10–60× (Supplementary Table 18); `bridge.py` with default parameters on the input given to MethPhaser; one local fix (two undefined debug counters stopped every run with an error; phasing logic unchanged)

Continued on next page

Supplementary Table 5 (continued)

Item	Version	Notes
LongHap[11]	0.2.0	nanopore, replicate 1, 10–60× (Supplementary Table 18); `-ont -min_base_quality 10`, as its documentation recommends for nanopore data, run per chromosome; SNVs with `-snvs_only`, with and without 5mC; SNVs, indels and SVs with 5mC from one VCF of the PEPPER calls and Sniffles2 deletions and insertions made sequence-resolved (inversions, duplications and breakends, 296–373 per coverage, omitted); 5mC from modkit 0.6.4 `pileup -combine-strands -cpg`
TinkerHap[12]	1.0.0	nanopore 10–60× and HiFi 10–50×, replicate 1 (Supplementary Table 18); default parameters, run per chromosome; phases indels by default and has no SNV-only mode, so SNV-only phasing used the call sets restricted to SNVs
HiPhase[13]	1.7.0	HiFi 10–50× (Supplementary Table 18); default parameters with `-threads 16 -ignore-read-groups`; phases indels by default and has no SNV-only mode, so SNV-only phasing used the call sets restricted to SNVs
HG002 PacBio HiFi reads	Revio, PacBio release 2023-10-31	ccs 7.0.0, lima 2.7.1, primrose 1.4.0; aligned with pbmm2 1.10.0 (48×); down-sampled with samtools 1.23.1 (seed 1) to 10–40×, 48× used as 50×; PEPPER-Margin-DeepVariant r0.8 `-hifi`, called separately at every coverage
LongPhase 1.0	v1.0	`-ont`; commit `fe8c1fdff11ba2a082d3 63a7ffcb5c923cacc316`
LongPhase 2	v2.1.1, commit `f9962dd`	Nanopore mode (`-pb` for HiFi); GNN correction with defaults

Supplementary Table 6: Selected applications and evaluations of LongPhase. Selected from works citing LongPhase (Europe PMC and OpenAlex, September 2026); the last column paraphrases the Methods of each work.

Work	Journal, year	Relation to LongPhase
Variant callers and pipelines		
ClairS [14]	Nat. Methods 2026	default germline phasing of tumour and normal alignments (`-whatshap_for_phasing` the alternative); haplotagging selectable
ClairS-TO [15]	Nat. Commun. 2025	default intermediate phasing and haplotagging of the tumour alignment (WhatsHap selectable)
Clair3-RNA [16]	Nat. Commun. 2025	phasing in the long-read RNA pipeline
Clair3 (accelerated) [17]	Bioinformatics 2026	selectable intermediate phasing and haplotagging (`-longphase_for_phasing`; WhatsHap the default)
Severus [18]	Nat. Biotechnol. 2025	one of the documented routes to the haplotagged BAMs it consumes (WhatsHap the other)
EPI2ME wf-human-variation [19]	ONT workflow	intermediate per-contig phasing (`phase_contig`); WhatsHap for final phasing and haplotagging
HiFi-somatic-WDL [20]	PacBio workflow	optional phasing tool
cuteHap [21]	Adv. Sci. 2026	read phasing before haplotype-aware SV calling
SpecImmune [22]	Adv. Sci. 2026	joint SNV–SV phasing of immune genes
Biological and clinical studies		
Fluctuating DNA methylation in cancer [23]	Nature 2025	read phasing (v1.7) and `modcall` for CpG methylation
Advanced cancer cohort (POG) [24]	Cell Genomics 2024	discussed as a phasing method for tumours
Pediatric epilepsy episignatures [25]	Nat. Commun. 2024	phasing of Clair3 and SV calls
GNAS transcripts in PHP1B [26]	PNAS 2025	phasing of Clair3 calls
3-hour genome for neonatal risk [27]	Genet. Med. Open 2024	real-time phasing during sequencing
Ultrarapid diagnostics in critical care [28]	Eur. J. Hum. Genet. 2025	read phasing (v1.0.7)
HYDIN in primary ciliary dyskinesia [29]	Eur. J. Hum. Genet. 2024	phasing of Clair3 calls
Antisense oligonucleotides in APBD [30]	Nucleic Acids Res. 2025	phasing of Clair3 calls
Native CpG methylation fine-mapping [31]	HGG Adv. 2026	phasing and haplotagging
Canadian national platform benchmark [32]	Front. Genet. 2026	phasing inside the Clair3 workflow
Pallas's cat genome [33]	NAR Genom. Bioinform. 2023	haplotagging for allele-specific methylation
Methods evaluated against LongPhase		
HiPhase [13]	Bioinformatics 2024	comparison method
Ralphi [34]	Genome Res. 2025	comparison method

Work	Journal, year	Relation to LongPhase
GCphase[35]	BMC Bioinformatics 2024	comparison method
Duet[36]	BMC Bioinformatics 2022	comparison method for SV phasing

Supplementary Table 7: Input call sets per coverage. Heterozygous (het.) and homozygous (hom.) calls entering the phasing runs; nanopore values at 10–20× are means (s.d.) over ten down-sampling replicates. Small variants from PEPPER-Margin-DeepVariant, SVs from Sniffles2 2.8.0, allele-specific 5mC sites from `longphase modcall`. SVs and 5mC were not called on the HiFi data. Homozygous SNVs include the PEPPER candidates genotyped homozygous reference (`refCall`).

Platform	Cov.	Het. SNV	Hom. SNV	Het. indel	Het. SV	Hom. SV	Het. 5mC
Nanopore	10×	2,340,598 (3,826)	3,265,243 (8,954)	283,612 (381)	14,866 (130)	7,684 (57)	283,932 (1,026)
Nanopore	12×	2,506,797 (2,922)	3,227,137 (8,307)	315,147 (511)	16,119 (69)	7,874 (33)	361,073 (2,274)
Nanopore	14×	2,586,505 (3,184)	3,173,111 (5,212)	332,537 (343)	16,707 (67)	8,008 (31)	408,058 (1,847)
Nanopore	16×	2,621,584 (2,784)	3,115,855 (8,105)	342,202 (505)	17,089 (68)	8,121 (17)	431,895 (976)
Nanopore	18×	2,636,024 (1,840)	3,051,563 (5,197)	347,703 (324)	17,326 (80)	8,201 (34)	444,081 (1,158)
Nanopore	20×	2,640,440 (2,408)	3,004,806 (5,806)	350,682 (328)	17,487 (43)	8,247 (32)	452,032 (1,079)
Nanopore	30×	2,636,000	2,902,347	356,240	17,800	8,473	474,512
Nanopore	40×	2,617,161	2,902,110	358,059	17,764	8,635	485,070
Nanopore	50×	2,598,083	2,914,706	360,142	17,602	8,754	493,774
Nanopore	60×	2,579,823	2,929,398	361,119	17,474	8,834	499,496
HiFi	10×	2,367,609	2,081,641	425,786	–	–	–
HiFi	20×	2,485,185	2,318,132	442,122	–	–	–
HiFi	30×	2,468,170	2,428,526	432,578	–	–	–
HiFi	40×	2,438,078	2,508,667	425,166	–	–	–
HiFi	50×	2,410,815	2,566,571	420,096	–	–	–

Supplementary Table 8: SNV-only phasing of HG002 nanopore R10.4.1 data by six configurations. Scored against the v5.0q VCF (chr1–22, no benchmark BED applied). Values at 10–20× are means (s.d.) over ten down-sampling replicates; 30–60×, one replicate. Phased SNVs are given as a percentage of the 2,398,880 heterozygous SNVs of the benchmark. Precision, phased SNV calls that match a benchmark heterozygous SNV as a percentage of all phased heterozygous SNV calls on chr1–22 (LongPhase 2 and WhatsHap only); F1, harmonic mean of precision and phased SNVs. Rate ratio, switch error rate divided by that of LongPhase 2 in the same down-sampling replicate. Margin and Ralphi were run at 10–20× only; GCphase was not included (Methods). Ralphi's phase blocks sum to 4.3–5.8 Gb, more than the length of the autosomes, so its block N50 is not comparable with that of the other tools.

Cov.	Phased SNV (%)	Precision (%)	F1 (%)	Switch errors	Switch error rate (%)	Rate ratio	Hamming (%)	Blocks	N50 (Mb)
LongPhase 2									
10×	77.84 (0.07)	89.63 (0.17)	83.32 (0.09)	1,503 (29)	0.081 (0.002)	–	5.39 (0.24)	7,040 (38)	0.84 (0.01)
12×	84.48 (0.06)	90.29 (0.11)	87.29 (0.07)	1,313 (50)	0.065 (0.002)	–	5.37 (0.38)	5,822 (54)	1.11 (0.02)
14×	87.88 (0.05)	90.70 (0.09)	89.27 (0.04)	1,120 (40)	0.053 (0.002)	–	4.89 (0.53)	5,193 (49)	1.29 (0.03)
16×	89.58 (0.03)	90.95 (0.17)	90.26 (0.09)	985 (27)	0.046 (0.001)	–	4.24 (0.23)	4,806 (46)	1.44 (0.02)
18×	90.50 (0.04)	91.30 (0.11)	90.90 (0.05)	896 (27)	0.041 (0.001)	–	3.69 (0.22)	4,530 (55)	1.54 (0.03)
20×	90.94 (0.02)	91.52 (0.16)	91.23 (0.08)	804 (39)	0.037 (0.002)	–	3.22 (0.35)	4,339 (41)	1.62 (0.02)
30×	91.49	92.08	91.79	648	0.030	–	1.85	3,742	1.93
40×	91.54	92.57	92.05	594	0.027	–	1.71	3,364	2.20
50×	91.45	92.80	92.12	588	0.027	–	1.54	3,160	2.53
60×	91.35	93.27	92.30	477	0.022	–	1.58	2,957	2.87
LongPhase 2 without GNN									
10×	78.10 (0.07)	–	–	2,259 (39)	0.121 (0.002)	1.50 (0.02)	7.04 (0.29)	6,056 (46)	0.92 (0.01)
12×	84.71 (0.06)	–	–	1,995 (60)	0.098 (0.003)	1.52 (0.04)	7.21 (0.42)	4,878 (39)	1.21 (0.02)
14×	88.07 (0.05)	–	–	1,706 (50)	0.081 (0.002)	1.52 (0.05)	6.43 (0.55)	4,314 (52)	1.40 (0.02)
16×	89.75 (0.03)	–	–	1,514 (40)	0.070 (0.002)	1.53 (0.03)	5.51 (0.21)	3,964 (43)	1.54 (0.03)
18×	90.65 (0.03)	–	–	1,374 (36)	0.063 (0.002)	1.53 (0.04)	4.75 (0.22)	3,740 (34)	1.62 (0.03)
20×	91.08 (0.02)	–	–	1,235 (45)	0.057 (0.002)	1.54 (0.04)	4.12 (0.38)	3,564 (35)	1.70 (0.02)
30×	91.61	–	–	1,004	0.046	1.55	2.40	3,007	1.97
40×	91.64	–	–	878	0.040	1.48	2.37	2,624	2.29
50×	91.55	–	–	883	0.040	1.50	1.86	2,389	2.57
60×	91.43	–	–	743	0.034	1.56	1.75	2,225	2.91
WhatsHap (SNVs only)									
10×	78.74 (0.07)	86.22 (0.12)	82.31 (0.07)	4,383 (65)	0.233 (0.003)	2.88 (0.06)	7.60 (0.39)	5,960 (49)	0.95 (0.01)
12×	85.35 (0.05)	87.09 (0.12)	86.21 (0.08)	4,229 (126)	0.207 (0.006)	3.19 (0.19)	7.99 (0.40)	4,897 (45)	1.24 (0.03)
14×	88.71 (0.05)	87.79 (0.10)	88.24 (0.06)	3,848 (127)	0.181 (0.006)	3.40 (0.10)	7.01 (0.42)	4,435 (47)	1.41 (0.02)
16×	90.37 (0.04)	88.30 (0.12)	89.32 (0.06)	3,561 (134)	0.165 (0.006)	3.58 (0.10)	6.37 (0.33)	4,169 (52)	1.53 (0.02)
18×	91.27 (0.03)	88.65 (0.08)	89.94 (0.04)	3,291 (134)	0.151 (0.006)	3.64 (0.17)	5.23 (0.39)	4,009 (34)	1.60 (0.04)
20×	91.68 (0.02)	89.10 (0.09)	90.38 (0.04)	3,067 (147)	0.140 (0.007)	3.79 (0.30)	4.40 (0.35)	3,904 (38)	1.65 (0.02)
30×	92.26	89.72	90.98	2,524	0.114	3.86	2.56	3,499	1.88
40×	92.36	90.45	91.39	2,139	0.097	3.57	3.05	3,208	2.15
50×	92.37	91.04	91.70	2,010	0.091	3.38	2.40	3,034	2.47

Cov.	Phased SNV (%)	Precision (%)	F1 (%)	Switch errors	Switch error rate (%)	Rate ratio	Hamming (%)	Blocks	N50 (Mb)
60×	92.34	91.52	91.93	1,785	0.081	3.70	2.75	2,912	2.72
HapCUT2									
10×	78.78 (0.09)	–	–	5,018 (108)	0.266 (0.006)	3.30 (0.08)	7.77 (0.39)	5,878 (55)	0.95 (0.01)
12×	85.41 (0.05)	–	–	4,948 (150)	0.242 (0.007)	3.73 (0.23)	8.06 (0.32)	4,837 (41)	1.25 (0.02)
14×	88.76 (0.07)	–	–	4,652 (104)	0.219 (0.005)	4.11 (0.13)	7.10 (0.52)	4,382 (48)	1.44 (0.03)
16×	90.44 (0.04)	–	–	4,344 (96)	0.201 (0.004)	4.37 (0.15)	6.13 (0.41)	4,122 (50)	1.54 (0.02)
18×	91.32 (0.05)	–	–	4,068 (121)	0.186 (0.006)	4.50 (0.17)	5.32 (0.29)	3,969 (39)	1.61 (0.04)
20×	91.70 (0.03)	–	–	3,825 (140)	0.174 (0.006)	4.73 (0.33)	4.27 (0.35)	3,872 (42)	1.66 (0.02)
30×	92.27	–	–	3,280	0.148	5.02	2.61	3,482	1.89
40×	92.37	–	–	2,916	0.132	4.87	2.90	3,203	2.16
50×	92.38	–	–	2,568	0.116	4.32	2.20	3,034	2.46
60×	92.34	–	–	2,309	0.104	4.79	2.11	2,911	2.72
Margin									
10×	78.53 (0.07)	–	–	1,286 (31)	0.069 (0.002)	0.86 (0.03)	0.47 (0.04)	18,177 (100)	0.26 (0.00)
12×	85.13 (0.06)	–	–	1,347 (32)	0.066 (0.002)	1.03 (0.05)	0.62 (0.10)	13,654 (88)	0.40 (0.01)
14×	88.47 (0.05)	–	–	1,331 (46)	0.063 (0.002)	1.19 (0.05)	0.72 (0.05)	11,181 (69)	0.52 (0.01)
16×	90.14 (0.03)	–	–	1,302 (49)	0.061 (0.002)	1.32 (0.05)	0.79 (0.11)	9,738 (58)	0.62 (0.01)
18×	91.04 (0.03)	–	–	1,281 (41)	0.059 (0.002)	1.43 (0.05)	0.87 (0.10)	8,818 (49)	0.71 (0.00)
20×	91.47 (0.02)	–	–	1,254 (60)	0.057 (0.003)	1.56 (0.12)	0.93 (0.11)	8,129 (46)	0.78 (0.01)
Ralphi									
10×	78.52 (0.08)	–	–	3,784 (104)	0.201 (0.006)	2.50 (0.08)	7.48 (0.27)	5,692 (49)	5.31 (7.10)
12×	85.14 (0.06)	–	–	3,694 (164)	0.181 (0.008)	2.80 (0.19)	8.32 (0.53)	4,536 (34)	7.97 (8.42)
14×	88.50 (0.06)	–	–	3,452 (102)	0.163 (0.005)	3.06 (0.12)	7.71 (0.63)	4,000 (58)	25.31 (16.84)
16×	90.17 (0.04)	–	–	3,176 (104)	0.147 (0.005)	3.20 (0.12)	7.60 (0.52)	3,681 (36)	30.67 (12.45)
18×	91.07 (0.03)	–	–	2,979 (163)	0.137 (0.007)	3.31 (0.21)	7.14 (0.51)	3,463 (30)	39.14 (14.83)
20×	91.50 (0.02)	–	–	2,874 (96)	0.131 (0.004)	3.56 (0.25)	6.56 (0.53)	3,314 (37)	47.11 (6.72)

Supplementary Table 9: The same phased VCFs scored against GIAB v4.2.1 and the T2T-HG002-derived v5.0q benchmark. Switch errors on SNVs; Hamming distance over SNVs and, where they are phased, indels. All LongPhase 2 runs with GNN correction, phasing SNVs alone or with indels (+indel) or indels and SVs (+indel+SV). Values at 10–20× are means (s.d.) over ten replicates. Phased indels as a percentage of benchmark heterozygous indels (v5.0q).

Configuration	Cov.	Switch errors v4.2.1	Switch errors v5.0q	Hamming v4.2.1 (%)	Hamming v5.0q (%)	Phased indels (%)
LongPhase 2, SNVs	10×	1,839 (28)	1,503 (29)	5.37 (0.23)	5.39 (0.24)	–
	20×	1,663 (31)	804 (39)	3.14 (0.36)	3.22 (0.35)	–
	30×	1,641	648	1.83	1.85	–
	40×	1,667	594	1.64	1.71	–
	50×	1,671	588	1.49	1.54	–
	60×	1,678	477	1.58	1.58	–
LongPhase 2, +indels	10×	2,000 (26)	1,708 (29)	6.95 (0.40)	6.98 (0.40)	30.33 (0.05)
	20×	1,818 (33)	981 (29)	5.83 (0.45)	5.84 (0.43)	40.99 (0.03)
	30×	1,789	806	4.77	4.96	43.35
	40×	1,766	714	4.32	4.49	44.36
	50×	1,768	671	3.27	3.43	44.99
	60×	1,771	588	3.61	3.56	45.32
LongPhase 2, +indels+SVs	10×	1,996 (24)	1,702 (30)	6.94 (0.39)	6.97 (0.40)	30.33 (0.05)
	20×	1,817 (37)	979 (28)	6.03 (0.50)	6.04 (0.48)	40.98 (0.03)
	30×	1,784	820	5.01	5.11	43.34
	40×	1,773	732	4.26	4.48	44.35
	50×	1,775	674	3.32	3.41	44.98
	60×	1,780	579	3.77	3.70	45.31
WhatsHap, SNVs only	10×	2,773 (47)	4,383 (65)	7.44 (0.37)	7.60 (0.39)	–
	20×	1,972 (29)	3,067 (147)	4.21 (0.38)	4.40 (0.35)	–
	30×	1,850	2,524	2.45	2.56	–
	40×	1,810	2,139	2.92	3.05	–
	50×	1,811	2,010	2.23	2.40	–
	60×	1,801	1,785	2.61	2.75	–
WhatsHap, SNVs + indels	10×	3,538 (48)	5,174 (78)	12.00 (0.34)	12.16 (0.36)	32.86 (0.03)
	20×	2,580 (32)	3,683 (122)	11.34 (0.59)	11.45 (0.57)	44.31 (0.04)
	30×	2,345	3,027	9.67	9.71	47.14
	40×	2,221	2,572	10.43	10.40	48.52
	50×	2,171	2,346	8.32	8.46	49.36

Configuration	Cov.	Switch errors v4.2.1	Switch errors v5.0q	Hamming v4.2.1 (%)	Hamming v5.0q (%)	Phased indels (%)
	60×	2,156	2,139	9.20	9.29	49.84
HapCUT2	10×	3,050 (33)	5,018 (108)	7.59 (0.38)	7.77 (0.39)	–
	20×	2,346 (40)	3,825 (140)	4.09 (0.35)	4.27 (0.35)	–
	30×	2,202	3,280	2.46	2.61	–
	40×	2,171	2,916	2.72	2.90	–
	50×	2,133	2,568	2.03	2.20	–
	60×	2,088	2,309	2.02	2.11	–

Supplementary Table 10: Co-phasing configurations of LongPhase 2, without and with GNN correction. Scored against v5.0q: switch errors on SNVs, Hamming distance over SNVs and, where they are phased, indels; phased indels as a percentage of benchmark heterozygous indels. Means (s.d.) over ten replicates at 10 and 20×; one replicate at 30 and 60×. Every coverage is in Supplementary Data 1.

Classes	GNN	Cov.	Phased indel (%)	Switch errors	Switch error rate (%)	Hamming (%)	N50 (Mb)
SNV	no	10×	–	2,259 (39)	0.121 (0.002)	7.04 (0.29)	0.92 (0.01)
		20×	–	1,235 (45)	0.057 (0.002)	4.12 (0.38)	1.70 (0.02)
		30×	–	1,004	0.046	2.40	1.97
		60×	–	743	0.034	1.75	2.91
	yes	10×	–	1,503 (29)	0.081 (0.002)	5.39 (0.24)	0.84 (0.01)
		20×	–	804 (39)	0.037 (0.002)	3.22 (0.35)	1.62 (0.02)
		30×	–	648	0.030	1.85	1.93
		60×	–	477	0.022	1.58	2.87
+SV	no	10×	–	2,292 (47)	0.123 (0.002)	7.23 (0.24)	0.93 (0.01)
		20×	–	1,262 (44)	0.058 (0.002)	4.44 (0.35)	1.72 (0.01)
		30×	–	1,031	0.047	2.64	1.98
		60×	–	745	0.034	2.09	2.93
	yes	10×	–	1,512 (32)	0.081 (0.002)	5.49 (0.22)	0.85 (0.01)
		20×	–	818 (41)	0.038 (0.002)	3.40 (0.30)	1.63 (0.02)
		30×	–	666	0.030	1.83	1.92
		60×	–	485	0.022	1.73	2.90
+5mC	no	10×	–	2,247 (45)	0.120 (0.002)	7.13 (0.33)	0.94 (0.01)
		20×	–	1,228 (41)	0.056 (0.002)	4.38 (0.52)	1.84 (0.03)
		30×	–	1,015	0.046	2.53	2.12
		60×	–	736	0.034	1.82	3.14
	yes	10×	–	1,522 (33)	0.082 (0.002)	5.48 (0.29)	0.86 (0.01)
		20×	–	806 (35)	0.037 (0.002)	3.36 (0.35)	1.75 (0.02)
		30×	–	658	0.030	2.05	2.08
		60×	–	485	0.022	1.60	3.09
+indel	no	10×	30.79 (0.04)	2,913 (52)	0.156 (0.003)	10.87 (0.39)	1.18 (0.02)
		20×	41.25 (0.03)	1,646 (47)	0.075 (0.002)	9.50 (0.38)	2.45 (0.05)
		30×	43.55	1,316	0.060	8.02	2.96
		60×	45.48	953	0.043	6.38	4.35
	yes	10×	30.33 (0.05)	1,708 (29)	0.092 (0.002)	6.98 (0.40)	0.98 (0.02)
		20×	40.99 (0.03)	981 (29)	0.045 (0.001)	5.84 (0.43)	2.03 (0.05)
		30×	43.35	806	0.037	4.96	2.48
		60×	45.32	588	0.027	3.56	3.74
+indel+SV	no	10×	30.78 (0.04)	2,934 (51)	0.157 (0.003)	11.00 (0.39)	1.19 (0.02)
		20×	41.24 (0.03)	1,657 (51)	0.076 (0.002)	9.76 (0.43)	2.48 (0.04)
		30×	43.54	1,323	0.060	8.45	2.99
		60×	45.48	952	0.043	6.51	4.37
	yes	10×	30.33 (0.05)	1,702 (30)	0.091 (0.002)	6.97 (0.40)	0.98 (0.02)
		20×	40.98 (0.03)	979 (28)	0.045 (0.001)	6.04 (0.48)	2.04 (0.05)
		30×	43.34	820	0.037	5.11	2.44
		60×	45.31	579	0.026	3.70	3.75
+indel+5mC	no	10×	30.79 (0.05)	2,877 (48)	0.154 (0.003)	10.98 (0.37)	1.21 (0.02)
		20×	41.26 (0.03)	1,596 (46)	0.073 (0.002)	9.70 (0.55)	2.70 (0.08)
		30×	43.55	1,317	0.060	8.01	3.19
		60×	45.49	937	0.043	7.33	5.03
	yes	10×	30.40 (0.05)	1,730 (25)	0.093 (0.001)	7.20 (0.46)	1.00 (0.01)
		20×	41.05 (0.04)	961 (26)	0.044 (0.001)	6.00 (0.54)	2.23 (0.05)
		30×	43.40	827	0.038	5.33	2.69
		60×	45.37	594	0.027	3.67	4.14
all four	no	10×	30.79 (0.05)	2,897 (49)	0.155 (0.003)	11.08 (0.43)	1.21 (0.02)
		20×	41.26 (0.03)	1,610 (52)	0.074 (0.002)	9.88 (0.62)	2.74 (0.07)

Classes	GNN	Cov.	Phased indel (%)	Switch errors	Switch error rate (%)	Hamming (%)	N50 (Mb)
		30×	43.54	1,316	0.060	8.33	3.19
		60×	45.48	932	0.043	7.23	5.10
	yes	10×	30.40 (0.05)	1,721 (29)	0.092 (0.002)	7.16 (0.49)	1.01 (0.02)
		20×	41.04 (0.03)	961 (28)	0.044 (0.001)	6.08 (0.52)	2.24 (0.06)
		30×	43.39	840	0.038	5.49	2.65
		60×	45.36	580	0.026	3.34	4.14

Supplementary Table 13: Location of SNV switch errors by truth set and benchmark region. **a**, switch errors by truth set and region; **b**, by genomic stratum within the v5.0q benchmark regions; **c**, switch errors shared by the three tools. SNV-only phasing of HG002 nanopore R10.4.1 data; LongPhase 2 with GNN correction. A variant pair is assigned to a region when both of its variants lie in it, so region counts do not sum to the total. *Shared*, inside both sets of benchmark regions; *v5.0q only*, inside the v5.0q regions only; *outside*, outside both. Means over ten replicates at 10 and 20×; one replicate at 30 and 60×. Rates are switch errors per assessed SNV pair. Per-run values for every coverage, region and stratum are in Supplementary Data 1.

		v4.2.1		v5.0q					
Tool	Cov.	All	Shared	All	Inside v5.0q	Rate inside v5.0q (%)	Shared	v5.0q only	Outside (rate, %)
LongPhase 2	10×	1,839	1,464	1,503	950	0.0534	819	41	416 (1.14)
	20×	1,663	1,175	804	270	0.0130	195	29	429 (0.92)
	30×	1,641	1,133	648	120	0.0057	63	26	424 (0.92)
	60×	1,678	1,180	477	71	0.0034	32	20	328 (0.76)
WhatsHap	10×	2,773	2,235	4,383	1,992	0.1112	1,483	266	2,030 (4.63)
	20×	1,972	1,372	3,067	723	0.0347	340	240	2,065 (3.75)
	30×	1,850	1,238	2,524	444	0.0211	134	205	1,829 (3.34)
	60×	1,801	1,245	1,785	235	0.0112	63	88	1,337 (2.61)
HapCUT2	10×	3,050	2,432	5,018	2,445	0.1364	1,686	322	2,175 (4.93)
	20×	2,346	1,633	3,825	1,264	0.0606	613	315	2,262 (4.10)
	30×	2,202	1,492	3,280	937	0.0446	393	268	2,053 (3.74)
	60×	2,088	1,435	2,309	676	0.0321	273	148	1,430 (2.78)

b Within the v5.0q benchmark regions, by GIAB v3.6 stratum

Tool	Cov.	Segmental duplication	Low mappability or seg. dup.	Tandem repeat	Homopolymer	Satellite	Not difficult
LongPhase 2	10×	52	56	5	1	2	612
	20×	36	38	3	0	2	137
	30×	31	31	0	0	0	46
	60×	24	25	0	0	0	18
WhatsHap	10×	264	279	33	6	12	1,079
	20×	232	239	29	5	10	240
	30×	202	205	16	5	3	91
	60×	80	82	15	0	0	38
HapCUT2	10×	292	308	54	19	12	1,040
	20×	275	283	45	21	10	244
	30×	247	254	36	19	6	100
	60×	120	124	25	21	3	40

c Switch errors at identical SNV pairs across tools, replicate 1

Truth	Cov.	LongPhase 2	WhatsHap	HapCUT2	All three	All three (% of LongPhase 2)
v4.2.1	10×	1,852	2,775	3,041	1,162	63
	20×	1,698	1,922	2,292	1,423	84
	30×	1,641	1,850	2,202	1,483	90
	60×	1,678	1,801	2,088	1,555	93
v5.0q	10×	1,483	4,258	4,920	539	36
	20×	893	2,735	3,552	288	32
	30×	648	2,524	3,280	237	37
	60×	477	1,785	2,309	182	38

Supplementary Table 11: SNV-only phasing of HG002 PacBio HiFi (Revio) data. One replicate per coverage, scored against v5.0q. The network was trained on nanopore data only and applied without retraining. Variants were called separately at every coverage (Methods). HiPhase has no SNV-only option and was given the call set restricted to SNVs. Precision, share of the phased heterozygous SNV calls on chr1–22 that match a benchmark heterozygous SNV (Fig. 4d); –, not computed.

Tool	Cov.	Phased SNV (%)	Precision (%)	Switch errors	Switch error rate (%)	Hamming (%)	Blocks	N50 (kb)
LongPhase 2	10×	87.01	92.16	1,020	0.049	0.73	15,924	341
	20×	91.72	93.08	878	0.040	1.13	12,199	488
	30×	91.90	93.70	829	0.038	0.81	11,181	540
	40×	91.87	94.43	815	0.037	0.77	10,569	573
	50×	91.56	94.96	714	0.033	0.82	10,237	584
LongPhase 2 without GNN	10×	87.12	–	1,328	0.064	0.96	15,708	346
	20×	91.84	–	1,268	0.058	1.28	11,980	492
	30×	92.02	–	1,137	0.052	1.02	10,926	545
	40×	91.98	–	1,153	0.052	0.98	10,347	580
	50×	91.66	–	988	0.045	1.10	10,010	588
WhatsHap (SNVs only)	10×	87.43	90.31	2,655	0.128	1.03	15,804	340
	20×	92.20	91.26	2,504	0.114	1.21	12,090	478
	30×	92.40	92.43	2,331	0.106	0.92	11,082	525
	40×	92.37	93.74	1,954	0.089	0.87	10,512	561
	50×	92.05	94.61	1,687	0.077	0.68	10,186	577
HapCUT2	10×	87.47	–	3,174	0.152	1.06	15,779	340
	20×	92.25	–	3,071	0.140	1.13	12,084	480
	30×	92.45	–	2,806	0.127	1.00	11,088	527
	40×	92.41	–	2,345	0.106	0.97	10,509	562
	50×	92.08	–	2,116	0.096	0.75	10,186	577
HiPhase	10×	87.20	91.05	1,768	0.085	0.80	15,715	343
	20×	91.95	91.60	1,662	0.076	1.03	12,036	490
	30×	92.19	92.28	1,499	0.068	0.86	11,028	538
	40×	92.19	93.26	1,378	0.063	0.87	10,433	567
	50×	91.88	93.89	1,256	0.057	0.85	10,106	584

Supplementary Table 14: Phase accuracy of indels against v5.0q. HG002 nanopore R10.4.1 data, co-phasing runs. Indel-pair switch errors are counted over consecutive variant pairs of an intersected block that include at least one indel; the indel Hamming distance uses the block orientation chosen on all variants. Truth VCF used in full; the last column restricts the indel-pair rate to the v5.0q benchmark regions. Means (s.d.) over ten replicates at 10 and 20×; one replicate at 30 and 60×. Per-run values are in Supplementary Data 1.

Configuration	Cov.	Phased indels	Indel-pair switch errors	Indel-pair rate (%)	SNV+indel rate (%)	Indel Hamming (%)	Indel-pair rate, inside v5.0q (%)
LongPhase 2, +indel	10×	156,680 (239)	296 (26)	0.100 (0.009)	0.090 (0.001)	6.92 (0.40)	0.080 (0.008)
	20×	211,702 (155)	205 (10)	0.052 (0.002)	0.044 (0.001)	5.80 (0.40)	0.031 (0.002)
	30×	223,926	179	0.043	0.037	4.99	0.023
	60×	234,102	126	0.029	0.027	3.61	0.015
same, no GNN	10×	159,017 (231)	495 (31)	0.165 (0.010)	0.150 (0.003)	10.83 (0.41)	0.133 (0.009)
	20×	213,062 (161)	305 (10)	0.076 (0.002)	0.073 (0.002)	9.49 (0.34)	0.050 (0.003)
	30×	224,926	236	0.056	0.058	8.08	0.031
	60×	234,940	183	0.042	0.042	6.40	0.022
LongPhase 2, four classes	10×	157,013 (256)	308 (26)	0.104 (0.009)	0.090 (0.002)	7.08 (0.48)	0.085 (0.007)
	20×	212,004 (176)	199 (10)	0.050 (0.003)	0.043 (0.001)	6.03 (0.49)	0.030 (0.003)
	30×	224,123	173	0.041	0.038	5.56	0.022
	60×	234,304	129	0.029	0.026	3.47	0.015
WhatsHap, +indel	10×	169,742 (173)	2,685 (57)	0.840 (0.018)	0.350 (0.005)	12.54 (0.32)	0.734 (0.014)
	20×	228,884 (187)	3,379 (55)	0.788 (0.013)	0.272 (0.006)	11.91 (0.62)	0.700 (0.013)
	30×	243,513	3,764	0.827	0.262	10.37	0.751
	60×	257,426	4,377	0.911	0.253	9.81	0.858

Supplementary Table 15: SNV-only phasing on the chromosomes held out from GNN training (chr17, chr21 and chr22). HG002 nanopore R10.4.1 data; the phased VCFs of the genome-wide analysis scored with `longphase compare -regions=chr17,chr21,chr22` against each truth VCF in full. Switch errors of LongPhase 2 as mean (s.d.); switch error rates in %. *Removed*, switch errors removed by GNN correction relative to LongPhase 2 without it; *cost*, phased SNVs withheld per removed switch error (the deployment threshold was set on chr15 and chr16 to a cost of 6.35; Supplementary Method 4). Means over ten replicates at 10–20×; one replicate at 30–60×. Margin and Ralphi were run at 10–20× only.

		LongPhase 2		Switch error rate (%)					GNN correction	
Truth	Cov.	Switch errors	Rate (%)	without GNN	WhatsHap	HapCUT2	Margin	Ralphi	Removed (%)	Cost
v5.0q	10×	123 (17)	0.124	0.163	0.360	0.416	0.099	0.347	24.0	9.5
	12×	111 (15)	0.102	0.133	0.339	0.390	0.095	0.322	23.5	9.3
	14×	91 (11)	0.080	0.106	0.288	0.334	0.089	0.282	24.5	9.6
	16×	88 (12)	0.076	0.099	0.258	0.316	0.083	0.248	23.6	8.9
	18×	82 (7)	0.070	0.089	0.243	0.303	0.081	0.249	21.5	9.1
	20×	69 (14)	0.058	0.076	0.233	0.297	0.076	0.243	23.4	8.8
	30×	82	0.069	0.081	0.196	0.212	–	–	14.6	12.2
	40×	78	0.066	0.080	0.179	0.187	–	–	17.9	7.5
	50×	62	0.052	0.061	0.152	0.182	–	–	13.9	13.8
	60×	37	0.031	0.039	0.144	0.154	–	–	19.6	14.6
v4.2.1	10×	135 (9)	0.151	0.178	0.215	0.225	0.080	0.195	15.7	9.1
	12×	130 (7)	0.133	0.153	0.180	0.193	0.081	0.172	13.2	8.6
	14×	128 (3)	0.125	0.138	0.166	0.176	0.085	0.155	9.7	9.7
	16×	122 (7)	0.117	0.128	0.146	0.159	0.086	0.142	8.5	9.6
	18×	121 (5)	0.114	0.122	0.138	0.151	0.092	0.137	6.1	9.9
	20×	120 (6)	0.112	0.118	0.134	0.147	0.093	0.132	4.7	10.4
	30×	119	0.111	0.114	0.128	0.140	–	–	2.5	13.0
	40×	110	0.102	0.105	0.118	0.128	–	–	2.7	14.7
	50×	117	0.109	0.109	0.117	0.125	–	–	0.0	–
	60×	117	0.109	0.110	0.118	0.126	–	–	0.8	27.0

Supplementary Table 12: SNVs unphased by GNN correction, classified by the T2T-HG002 v1.1 assembly. Nanopore 60×, replicate 1, SNV-only phasing. "Aligned" denotes sites within regions where both assembled haplotypes align one-to-one to GRCh38 (`GRCh38_HG2-T2TQ100-V1.1_dipcall-z2k.dip.bed`). For these sites, the table also records whether the matching dipcall VCF contains a variant at the same start coordinate; alleles were not compared. All chromosomes, including chrX and chrY, unless stated. The background is a random sample of 50,000 SNVs phased before GNN correction, including sites that GNN correction later unphased. Top, overall classes; bottom, share outside the 1:1 alignment per chromosome. v5.0q is derived from the same assembly, so this analysis extends coverage rather than providing an independent truth set.

Class	SNVs	Share (%)
not aligned 1:1 (unassessed)	55,158	88.64
aligned, no assembly variant record at the same coordinate	3,843	6.18
aligned, assembly variant record at the same coordinate	3,229	5.19
total	62,230	100.00
aligned 1:1, unphased by GNN correction	7,072 / 62,230	11.36
aligned 1:1, phased before GNN correction (sample)	46,318 / 50,000	92.64
autosomes only, unphased by GNN correction	5,208 / 57,185	9.11
autosomes only, phased before GNN correction (sample)	46,116 / 49,622	92.93
inside the v5.0q benchmark BED	434	–
of which benchmark variant	296	68.20

Chromosome	Phased before GNN correction		Unphased by GNN correction		Ratio
	n	not aligned (%)	n	not aligned (%)	
chr1	3,787	7.05	4,040	90.17	12.8
chr2	3,713	3.42	1,962	94.50	27.6
chr3	3,467	4.99	2,819	80.06	16.0
chr4	3,355	3.37	1,342	78.61	23.3
chr5	3,184	2.73	1,465	61.57	22.6
chr6	3,135	2.90	788	90.36	31.2
chr7	2,949	5.49	3,088	80.73	14.7
chr8	2,707	5.13	2,337	91.36	17.8
chr9	2,342	12.34	1,809	95.69	7.8
chr10	2,478	4.16	1,905	91.18	21.9
chr11	2,639	13.19	10,617	98.92	7.5
chr12	2,420	9.13	4,663	99.55	10.9
chr13	1,801	6.11	2,866	96.51	15.8
chr14	1,590	2.01	310	43.23	21.5
chr15	1,482	8.70	1,456	79.88	9.2
chr16	1,496	8.29	2,907	84.14	10.1
chr17	1,512	11.11	1,611	71.69	6.5
chr18	1,679	24.84	8,029	98.06	3.9
chr19	1,130	0.09	133	53.38	593.1
chr20	1,219	11.16	2,007	90.18	8.1
chr21	781	20.23	642	90.50	4.5
chr22	756	14.68	389	80.98	5.5
chrX	263	44.11	4,456	68.85	1.6
chrY	115	52.17	589	19.19	0.4

Supplementary Table 16: Runtime and peak memory of SNV-only phasing. HG002 nanopore R10.4.1 data (top) and PacBio HiFi data (bottom), replicate 1 at every coverage, on the workstation described in Methods (Supplementary Fig. 10). Wall-clock time and CPU time (user plus system) in minutes and peak resident memory in GiB, from `/usr/bin/time`. LongPhase 2 includes GNN correction, which runs within the `phase` command. LongPhase 2 was timed on one thread at 10 and 60× only (–, no value); HapCUT2 at 30 and 40× was re-timed (Methods). On HiFi data, LongPhase 2 at every coverage and WhatsHap at 10 and 20× were re-timed with the alignment file already in the page cache, which shortens wall-clock time but leaves CPU time largely unchanged; CPU time is therefore the measure to compare between platforms (Methods). HapCUT2 includes fragment extraction.

	LongPhase 2, 24 threads			LongPhase 2, 1 thread			WhatsHap			HapCUT2		
Coverage	Wall	CPU	Memory	Wall	CPU	Memory	Wall	CPU	Memory	Wall	CPU	Memory
Nanopore R10.4.1												
10×	3.5	12.3	19.3	8.3	8.4	6.3	25.2	25.1	1.5	24.1	24.1	6.6
12×	4.0	13.1	22.1	–	–	–	37.6	37.6	1.6	28.1	28.0	6.4
14×	4.5	13.5	22.2	–	–	–	49.0	49.0	1.8	35.3	35.2	8.3
16×	5.4	14.2	23.0	–	–	–	58.9	58.8	1.9	39.4	39.3	8.9
18×	5.9	14.9	23.8	–	–	–	74.1	73.9	2.1	42.1	41.9	9.6
20×	6.1	15.4	23.9	–	–	–	74.6	73.1	2.2	44.5	42.6	8.6
30×	10.4	19.2	25.9	–	–	–	90.7	90.4	2.6	59.4	59.0	11.3
40×	13.5	23.3	27.3	–	–	–	100.4	100.1	3.0	75.6	75.2	14.0
50×	16.9	27.5	27.6	–	–	–	107.6	107.1	3.5	85.7	84.5	16.7
60×	15.5	30.7	30.1	19.1	18.8	7.4	113.8	113.6	4.0	96.4	95.9	19.4
PacBio HiFi												
10×	0.9	6.0	16.7	–	–	–	23.9	23.8	1.5	19.5	19.4	7.1
20×	1.0	8.7	19.5	–	–	–	62.9	62.9	2.2	44.6	44.4	11.4
30×	1.2	11.2	20.9	–	–	–	90.5	86.6	2.7	44.7	44.5	14.3
40×	1.7	13.0	21.8	–	–	–	93.5	93.2	3.2	59.8	59.5	17.2
50×	2.4	14.3	22.6	–	–	–	98.4	98.2	3.7	59.5	59.1	19.6

Supplementary Table 17: Heterozygous SNV calls phased by LongPhase 2 only, by WhatsHap only, by both or by neither. HG002 nanopore R10.4.1 data, SNV-only phasing, replicate 1; single-base calls with genotype 0/1, identical in both VCFs, including chrX and chrY. Calls phased by WhatsHap only are split by the LongPhase 2 stage that left them unphased: the phasing graph before GNN correction or GNN correction. Benchmark status against the chr1–22 v5.0q VCF without a BED: het., same position, REF and ALT; other allele, another heterozygous record at the position; hom., homozygous; absent, no record; chrX/Y, outside the truth. **Top**, composition. **Middle**, benchmark het. SNVs among calls phased by one tool only, scored for that tool: *assessed*, in a block with at least two assessed variants; *all phased*, the tool's block-wise Hamming distance over all its phased SNVs; *fold*, inside divided by expected. The definitions of wrong phase, switch-error intervals and expected share are in Methods, "Analysis of unphased SNVs". **Bottom**, additional calls phased by WhatsHap: calls phased by WhatsHap only minus calls phased by LongPhase 2 only.

Cov.	Phased by	Calls	Het.	Other allele	Hom.	Absent	chrX/Y
10×	LongPhase 2 only	34,359	1,501	61	1,077	30,980	740
	WhatsHap only	150,889	23,706	1,132	14,058	107,495	4,498
	unphased by the LongPhase 2 graph	125,981	17,933	924	12,070	91,429	3,625
	unphased by GNN correction	24,908	5,773	208	1,988	16,066	873
	both tools	2,055,955	1,865,973	2,382	13,077	162,113	12,410
	neither tool	93,527	2,523	221	3,939	83,360	3,484
30×	LongPhase 2 only	47,788	1,501	126	1,188	43,074	1,899
	WhatsHap only	134,583	19,974	937	5,316	102,887	5,469
	unphased by the LongPhase 2 graph	111,718	17,819	779	4,128	84,701	4,291
	unphased by GNN correction	22,865	2,155	158	1,188	18,186	1,178
	both tools	2,352,426	2,193,330	2,518	8,390	133,487	14,701
	neither tool	101,203	2,115	193	2,605	90,666	5,624
60×	LongPhase 2 only	49,514	968	123	1,260	44,353	2,810
	WhatsHap only	124,038	24,669	813	4,736	87,390	6,430
	unphased by the LongPhase 2 graph	99,813	23,094	684	3,652	67,392	4,991
	unphased by GNN correction	24,225	1,575	129	1,084	19,998	1,439
	both tools	2,313,472	2,190,399	2,255	7,079	102,954	10,785
	neither tool	92,799	1,604	157	1,925	81,195	7,918

			Benchmark het. SNVs among them				The scored tool's switch-error intervals		
Cov.	Phased by	Scored tool	Assessed	Wrong	Wrong (%)	All phased (%)	Inside (%)	Expected (%)	Fold
10×	LongPhase 2 only	LongPhase 2	1,478	188	12.7	5.31	3.74	1.59	2.35
	WhatsHap only	WhatsHap	23,616	5,386	22.8	7.71	10.03	2.33	4.31
30×	LongPhase 2 only	LongPhase 2	1,489	222	14.9	1.85	3.77	0.93	4.07
	WhatsHap only	WhatsHap	19,946	3,231	16.2	2.56	12.29	1.52	8.07
60×	LongPhase 2 only	LongPhase 2	958	142	14.8	1.58	10.48	1.06	9.92
	WhatsHap only	WhatsHap	24,629	2,432	9.9	2.75	10.61	1.12	9.48

Cov.	Additional calls phased by WhatsHap	Het. (%)	Other allele	Hom.	Absent	chrX/Y
10×	116,530	22,205 (19)	1,071	12,981	76,515	3,758
30×	86,795	18,473 (21)	811	4,128	59,813	3,570
60×	74,524	23,701 (32)	690	3,476	43,037	3,620

Supplementary Table 18: Phasing compared with further phasers. HG002, replicate 1; nanopore R10.4.1 data at 10–60× in steps of 10× and PacBio HiFi data at 10–50×. Each block compares tools given the same reads, variant calls and evidence classes; within a block, LongPhase 2 is listed first. MethPhaser and HapBridge were given the SNV-only output of LongPhase 2 without GNN correction. HiPhase and TinkerHap have no SNV-only mode, so their SNV-only rows used call sets restricted to SNVs. LongHap with indels, SVs and 5mC was given deletions and insertions from Sniffles2 as sequence-resolved records. Switch error rates are over consecutive SNV pairs; Hamming distance includes indels where both files contain them, so it is comparable only within a block. Phased fractions are percentages of benchmark heterozygous SNVs or indels; –, not applicable. Source: Supplementary Data 1, sheets `Other_phasers`, `SNV_Detail`, `co-phase_Detail` and `MethPhaser`.

Tool	Cov.	Phased SNVs (%)	Phased indels (%)	Switch error rate (%)	Rate relative to LongPhase 2	Hamming (%)	N50 (Mb)
Nanopore R10.4.1, SNVs and 5mC							
LongPhase 2	10×	77.88	–	0.083	–	5.38	0.88
	20×	90.97	–	0.040	–	3.41	1.74
	30×	91.50	–	0.030	–	2.05	2.08
	40×	91.54	–	0.028	–	2.19	2.49
	50×	91.46	–	0.028	–	1.85	2.87
	60×	91.37	–	0.022	–	1.60	3.09
MethPhaser	10×	78.11	–	0.124	1.50	7.19	0.96
	20×	91.09	–	0.069	1.72	7.16	2.28
	30×	91.61	–	0.052	1.73	5.96	3.35
	40×	91.64	–	0.043	1.56	4.55	4.06
	50×	91.55	–	0.043	1.55	5.13	5.03
	60×	91.43	–	0.035	1.59	3.95	5.49
HapBridge	10×	78.11	–	0.122	1.48	6.93	0.94
	20×	91.09	–	0.060	1.49	3.99	1.70
	30×	91.61	–	0.046	1.53	2.58	1.98
	40×	91.64	–	0.040	1.46	2.39	2.44
	50×	91.55	–	0.040	1.46	2.11	3.07
	60×	91.43	–	0.033	1.51	2.50	4.18
LongHap	10×	59.88	–	0.180	2.18	4.35	0.41
	20×	87.26	–	0.181	4.49	17.99	2.56
	30×	90.09	–	0.158	5.28	19.99	4.35
	40×	90.44	–	0.141	5.12	19.01	4.93
	50×	90.51	–	0.126	4.56	19.90	5.25
	60×	90.50	–	0.110	4.96	18.44	5.40
LongHap, without 5mC	10×	59.87	–	0.175	2.12	4.09	0.40
	20×	87.25	–	0.158	3.92	10.50	1.39
	30×	90.08	–	0.136	4.52	8.81	1.79
	40×	90.43	–	0.123	4.44	10.13	2.02
	50×	90.51	–	0.109	3.96	9.95	2.35
	60×	90.49	–	0.096	4.35	10.06	2.55
Nanopore R10.4.1, SNVs and indels							
LongPhase 2	10×	77.78	30.38	0.091	–	7.02	0.98
	20×	90.90	40.97	0.048	–	6.15	2.07
	30×	91.47	43.35	0.037	–	4.96	2.48
	40×	91.50	44.36	0.033	–	4.49	2.93
	50×	91.41	44.99	0.031	–	3.43	3.23
	60×	91.31	45.32	0.027	–	3.56	3.74
WhatsHap	10×	78.78	32.90	0.270	2.98	12.47	1.28
	20×	91.73	44.28	0.157	3.23	10.58	2.42
	30×	92.27	47.14	0.137	3.72	9.71	2.98
	40×	92.36	48.52	0.116	3.57	10.40	3.39
	50×	92.38	49.36	0.106	3.46	8.46	3.68
	60×	92.34	49.84	0.097	3.60	9.29	4.20
HapCUT2	10×	78.83	33.84	0.315	3.47	12.79	1.37
	20×	91.80	46.47	0.196	4.06	12.90	2.88
	30×	92.32	49.34	0.177	4.81	11.65	3.45
	40×	92.41	50.69	0.157	4.81	11.21	4.05
	50×	92.41	51.57	0.137	4.46	11.45	4.37
	60×	92.37	52.04	0.126	4.69	13.48	5.01
TinkerHap	10×	78.89	33.98	0.332	3.66	8.14	0.80
	20×	91.84	46.61	0.229	4.73	8.91	1.48
	30×	92.34	49.46	0.197	5.34	9.72	1.96
	40×	92.43	50.78	0.170	5.23	10.82	2.27
	50×	92.45	51.64	0.162	5.30	11.86	2.67
	60×	92.41	52.11	0.151	5.64	13.16	3.13
Nanopore R10.4.1, SNVs, indels, SVs and 5mC							
LongPhase 2	10×	77.79	30.46	0.092	–	6.92	1.02

Tool	Cov.	Phased SNVs (%)	Phased indels (%)	Switch error rate (%)	Rate relative to LongPhase 2	Hamming (%)	N50 (Mb)
	20×	90.91	41.02	0.047	–	6.95	2.32
	30×	91.47	43.39	0.038	–	5.49	2.65
	40×	91.50	44.41	0.033	–	4.73	3.18
	50×	91.43	45.02	0.031	–	3.73	3.63
	60×	91.34	45.36	0.026	–	3.34	4.14
LongHap	10×	59.84	27.13	0.219	2.39	6.43	0.47
	20×	87.26	43.76	0.194	4.10	19.52	2.95
	30×	90.09	47.68	0.166	4.34	21.91	4.52
	40×	90.43	49.28	0.148	4.47	21.68	5.81
	50×	90.51	50.23	0.129	4.21	20.79	5.79
	60×	90.49	50.73	0.114	4.32	20.97	6.83
PacBio HiFi, SNVs							
LongPhase 2	10×	87.01	–	0.049	–	0.73	0.34
	20×	91.72	–	0.040	–	1.13	0.49
	30×	91.90	–	0.038	–	0.81	0.54
	40×	91.87	–	0.037	–	0.77	0.57
	50×	91.56	–	0.033	–	0.82	0.58
WhatsHap	10×	87.43	–	0.128	2.59	1.03	0.34
	20×	92.20	–	0.114	2.84	1.21	0.48
	30×	92.40	–	0.106	2.80	0.92	0.52
	40×	92.37	–	0.089	2.38	0.87	0.56
	50×	92.05	–	0.077	2.35	0.68	0.58
HapCUT2	10×	87.47	–	0.152	3.10	1.06	0.34
	20×	92.25	–	0.140	3.48	1.13	0.48
	30×	92.45	–	0.127	3.36	1.00	0.53
	40×	92.41	–	0.106	2.86	0.97	0.56
	50×	92.08	–	0.096	2.95	0.75	0.58
HiPhase	10×	87.20	–	0.085	1.73	0.80	0.34
	20×	91.95	–	0.076	1.89	1.03	0.49
	30×	92.19	–	0.068	1.80	0.86	0.54
	40×	92.19	–	0.063	1.69	0.87	0.57
	50×	91.88	–	0.057	1.75	0.85	0.58
TinkerHap	10×	87.61	–	0.169	3.44	1.22	0.34
	20×	92.33	–	0.146	3.64	1.03	0.48
	30×	92.52	–	0.130	3.45	0.98	0.52
	40×	92.48	–	0.109	2.93	1.00	0.56
	50×	92.14	–	0.095	2.92	0.97	0.58
PacBio HiFi, SNVs and indels							
LongPhase 2	10×	86.86	47.64	0.094	–	2.54	0.40
	20×	91.65	53.06	0.060	–	2.06	0.56
	30×	91.87	55.69	0.054	–	1.84	0.63
	40×	91.83	57.15	0.048	–	1.64	0.66
	50×	91.52	57.76	0.041	–	1.49	0.68
WhatsHap	10×	87.55	50.64	0.252	2.67	4.89	0.44
	20×	92.27	55.23	0.176	2.92	4.13	0.60
	30×	92.46	57.55	0.144	2.69	3.24	0.65
	40×	92.42	58.86	0.112	2.31	2.20	0.67
	50×	92.09	59.37	0.096	2.37	2.05	0.69
HapCUT2	10×	87.59	53.56	0.272	2.89	5.44	0.45
	20×	92.33	61.26	0.213	3.53	4.91	0.65
	30×	92.52	64.16	0.177	3.31	4.27	0.70
	40×	92.47	65.77	0.147	3.05	3.51	0.73
	50×	92.13	66.45	0.131	3.24	3.39	0.75
HiPhase	10×	87.40	52.85	0.164	1.75	4.15	0.45
	20×	92.13	60.48	0.112	1.86	3.43	0.65
	30×	92.34	63.38	0.091	1.71	2.60	0.71
	40×	92.33	65.00	0.079	1.64	2.46	0.76
	50×	92.02	65.68	0.070	1.74	2.11	0.78
TinkerHap	10×	87.65	53.68	0.283	3.00	5.36	0.44
	20×	92.38	61.34	0.217	3.61	4.75	0.63
	30×	92.56	64.24	0.181	3.38	3.94	0.69
	40×	92.51	65.85	0.153	3.18	3.85	0.72
	50×	92.17	66.52	0.133	3.29	3.44	0.75

Supplementary Table 19: GNN models trained with two label definitions and five seeds. Each model was trained on the same windows with the settings of Supplementary Table 4; labels matched the v5.0q benchmark to the calls by position (*position*) or by position, REF and ALT (*REF/ALT*). Selected epoch and held-out classification (error class, argmax) from training; B, onset-score threshold that gives on the validation chromosomes (chr15, chr16) the cost of the release v2.1 model at $B = 0.30$ (6.35 phased SNVs withheld per removed switch error). At that threshold: switch errors removed by GNN correction relative to phasing without it, and the cost, on the validation chromosomes, the held-out chromosomes (chr17, chr21, chr22) and the whole genome, over the 60 HG002 nanopore SNV-only runs at 10–20×, applied with the Python correction pipeline and scored with `longphase compare` against v5.0q. The released model (v2.1.1) is marked with an asterisk; its threshold was rounded to 0.35, and the values of the released C++ implementation are in Supplementary Method 4 and Supplementary Table 15.

			Held-out		Switch errors removed (%)			Cost	
Labels	Seed	Epoch	F_1	B	Valid.	Held-out	Genome	Held-out	Genome
Position	42*	30	0.347	0.340	50.8	23.9	34.7	9.5	8.2
Position	43	58	0.348	0.263	48.1	22.4	32.9	9.8	8.1
Position	44	37	0.345	0.311	49.6	22.6	33.9	10.2	8.1
Position	1337	27	0.355	0.343	50.3	23.2	34.0	9.7	8.1
Position	3407	34	0.349	0.336	49.7	23.0	33.6	9.4	7.8
REF/ALT	42	48	0.381	0.232	49.8	22.9	34.1	9.8	8.3
REF/ALT	43	19	0.382	0.346	50.5	21.9	33.3	9.4	8.2
REF/ALT	44	52	0.367	0.218	47.9	20.8	33.1	10.6	8.5
REF/ALT	1337	44	0.369	0.238	49.4	21.4	33.6	10.2	8.3
REF/ALT	3407	42	0.370	0.251	49.1	21.2	33.6	10.4	8.4
Release v2.1 model	42	44	0.340	0.30	48.5	21.2	33.2	10.5	8.3

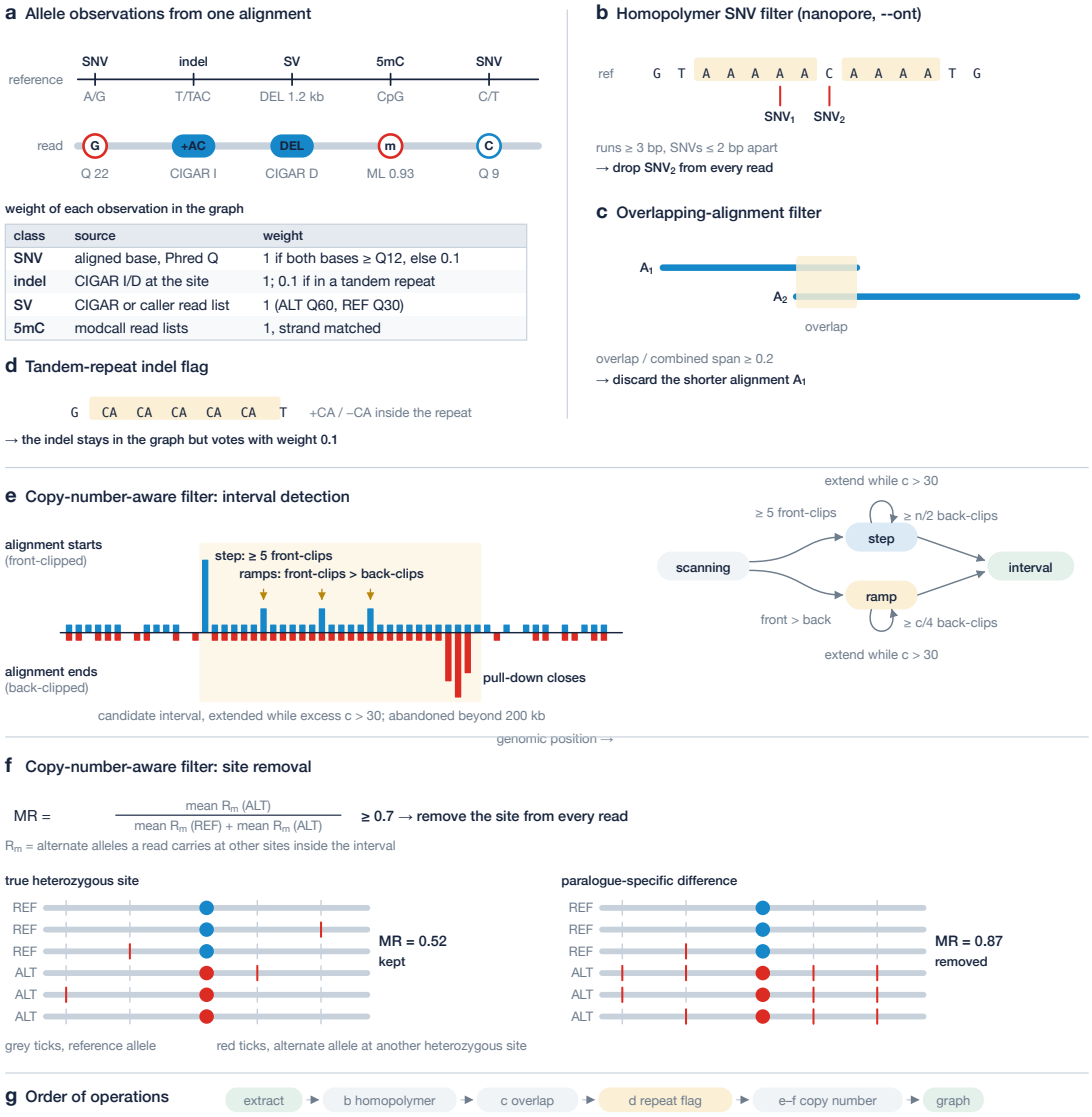
Supplementary Data 1. `supplementary.xlsx` contains the metrics behind Figs. 2–4, Supplementary Figs. 9 and 11–16 and Supplementary Tables 7–11, 13, 14, 17 and 18. Per-run metrics are indexed by coverage and down-sampling replicate. Blank cells denote runs that were not performed. The 14 sheets are described below.

The four `Unphased` sheets report replicate 1 at 10, 30 and 60×. They use `lp_only` for calls left unphased by LongPhase 2 only (phased by WhatsHap only), and `wh_only` for calls left unphased by WhatsHap only (phased by LongPhase 2 only). In `Unphased_composition`, `lp_only_phase` and `lp_only_gnn` divide `lp_only` into calls left unphased by the phasing graph before GNN correction and calls unphased by GNN correction, respectively.

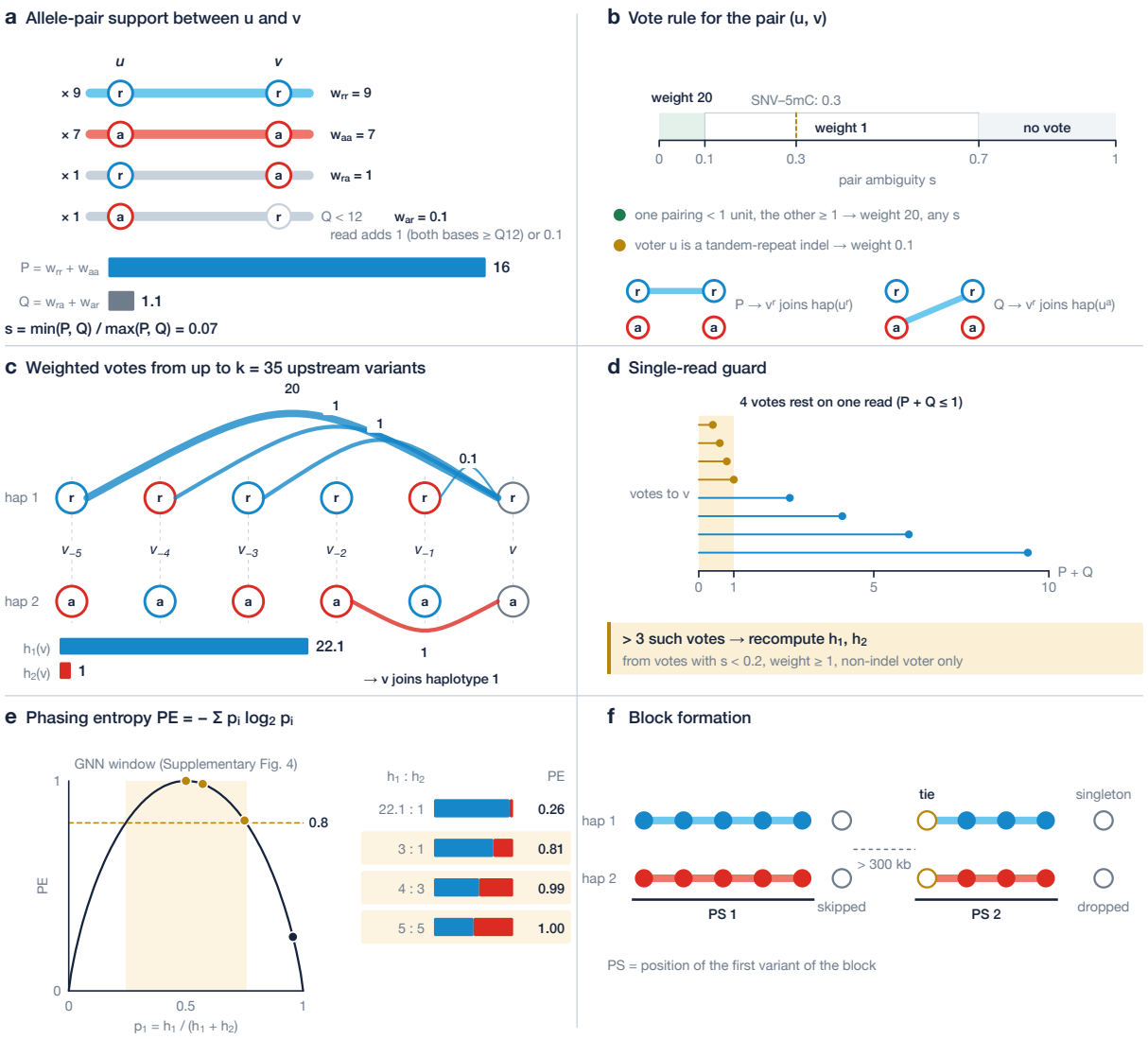
The code `both`, present in the composition, depth and sets sheets, denotes calls left unphased by both tools ("neither tool" in Supplementary Table 17). The code `background`, present in the composition and sets sheets, denotes calls phased by both tools ("both tools" in Supplementary Table 17).

Sheet	Contents
`Variant_Calling`	Input call sets.
`SNV_Detail`	SNV-only phasing by every tool on nanopore and HiFi data.
`co-phase_Detail`	The six co-phasing configurations of LongPhase 2 without (`longphase_v2.0.1`) and with (`longphase_v2.1.1`) GNN correction, and WhatsHap with indels.
`SNV_switch_by_region`	SNV switch errors and assessed pairs for SNV-only runs, by truth set, benchmark region and GIAB v3.6 stratum.
`SNV_switch_coincidence`	Switch errors at identical SNV pairs across tools (replicate 1).
`Indel_accuracy`	Indel and SNV switch errors, assessed pairs and Hamming distance for co-phasing runs, without and within the v5.0q benchmark regions.
`Unphased_composition`	Benchmark status of the calls left unphased, and phase errors of the tool that phased them (Fig. 4e and Supplementary Table 17).
`Unphased_intervals`	Co-location of these calls with switch-error intervals of the tool that phased them (Supplementary Table 17).
`Unphased_depth`	Read-depth histograms in 2× bins (Supplementary Fig. 12).
`Unphased_sets`	Per set: read-depth quartiles and share at the depth cap, phasing entropy (PE) of LongPhase 2 where it reports one, and enrichment in the other tool's switch-error intervals.
`Phased_calls`	Phased heterozygous SNV calls on chr1–22 for every SNV-only run of LongPhase 2 and WhatsHap, and on HiFi data also of HiPhase; the denominator of precision in Fig. 4b,d and Supplementary Tables 8 and 11.
`MethPhaser`	MethPhaser applied to the SNV-only output of LongPhase 2 without GNN correction, scored as in `SNV_Detail` (Fig. 2d and Supplementary Figs. 11 and 16).
`SV_5mC_phasing`	Heterozygous SVs (Sniffles2) and 5mC sites (`modcall`) of every co-phasing configuration with SVs or 5mC, and how many LongPhase 2 phased without (`longphase_v2.0.1`) and with (`longphase_v2.1.1`) GNN correction; all contigs, as in `Variant_Calling`.
`Other_phasers`	Replicate 1 of further phasers, run with default parameters or the documented workflow, scored as in `SNV_Detail`: LongHap (SNVs with and without 5mC, and all variant types with 5mC), HapBridge and TinkerHap on nanopore data; HiPhase and TinkerHap on HiFi data, with indels and on an SNV-only input VCF (neither has an SNV-only option); and LongPhase 2, WhatsHap and HapCUT2 with indels (Supplementary Table 18). The SNV-only HiPhase rows are also the HiPhase values of Figs. 2c,f and 4b, Supplementary Fig. 16e and Supplementary Table 11.

Supplementary Figures



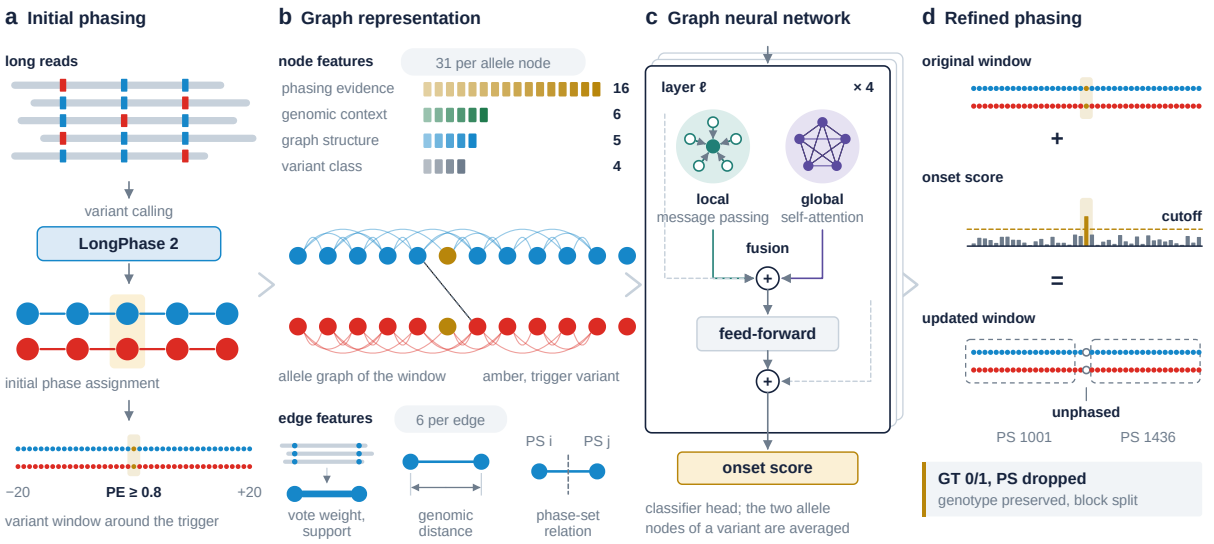
Supplementary Fig. 1: Read-level allele observations and pre-graph filters. **a**, Allele observations extracted from one alignment for each variant class, and the weight of each observation in the phasing graph (Supplementary Method 2). **b**, Homopolymer SNV filter (`--ont`): of two SNVs in homopolymers of length ≥ 3 that are ≤ 2 bp apart, the downstream SNV is removed. **c**, Overlapping-alignment filter for supplementary alignments of one read. **d**, Indels in tandem repeats remain in the graph and vote with weight 0.1. **e**, Clusters of clipped alignments define candidate intervals for the copy-number-aware filter (Supplementary Method 2). **f**, Within candidate intervals, heterozygous sites with MR ≥ 0.7 are removed from all reads; MR is the mean alternate-allele load of the reads carrying the alternate allele relative to the summed means of both read groups. **g**, Order in which the filters are applied.



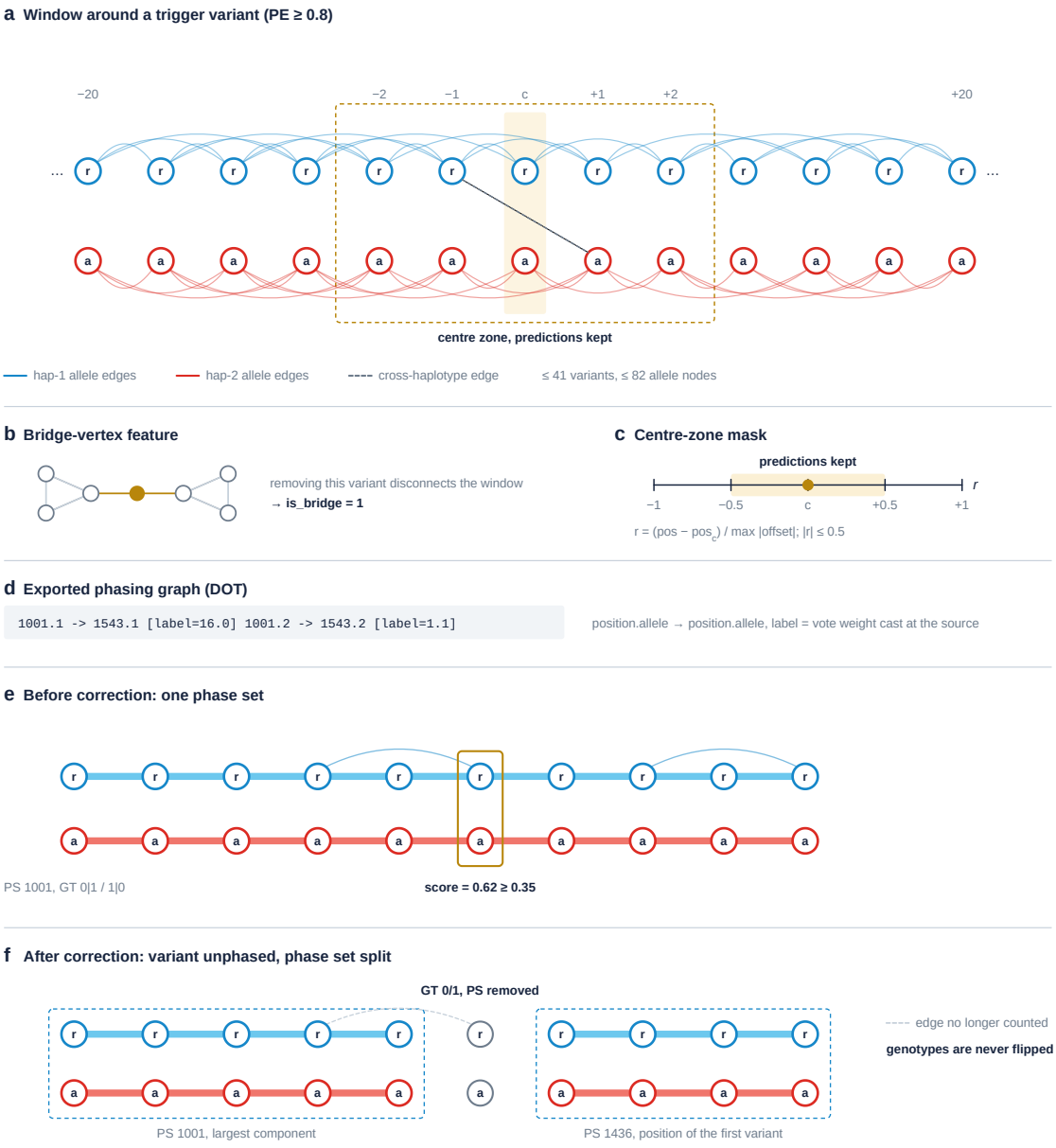
Supplementary Fig. 2: Pairwise allele support, weighted voting and phasing entropy. **a**, Reads spanning two variants u and v accumulate four weights; P and Q are the supports for the two possible pairings and $s = \min(P, Q)/\max(P, Q)$ is their ambiguity. **b**, Vote weight as a function of s; the rules are given in Methods. **c**, Variant v receives weighted votes from up to 35 upstream variants that share a read with it; line thickness encodes vote weight, and the totals h_1 and h_2 decide its haplotype. **d**, Single-read guard: when more than three votes rest on one read, h_1 and h_2 are recomputed from well-supported votes only. **e**, Phasing entropy (PE) of the vote totals as a function of the vote split (left) and for four examples (right); values ≥ 0.8 (shaded) open a window for the network (Supplementary Fig. 4). **f**, Block formation; the phase set (PS) is the position of the first variant of a block.



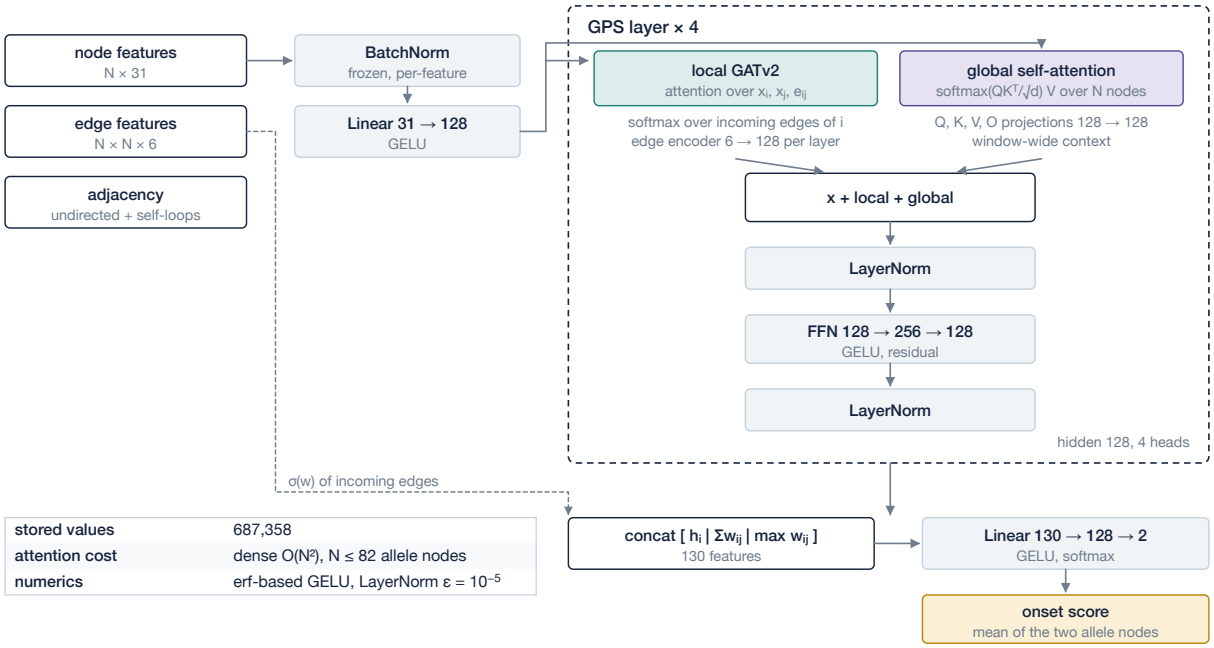
Supplementary Fig. 3: Read-based correction. **a**, Each read is assigned to a haplotype by a weighted majority over its phased alleles; reads with a majority ≤ 0.65 or fewer than two informative alleles remain untagged. **b**, The phase of each variant is re-derived from the tagged reads and retained only if the consistent fraction ρ exceeds 0.75; otherwise the variant is unphased.



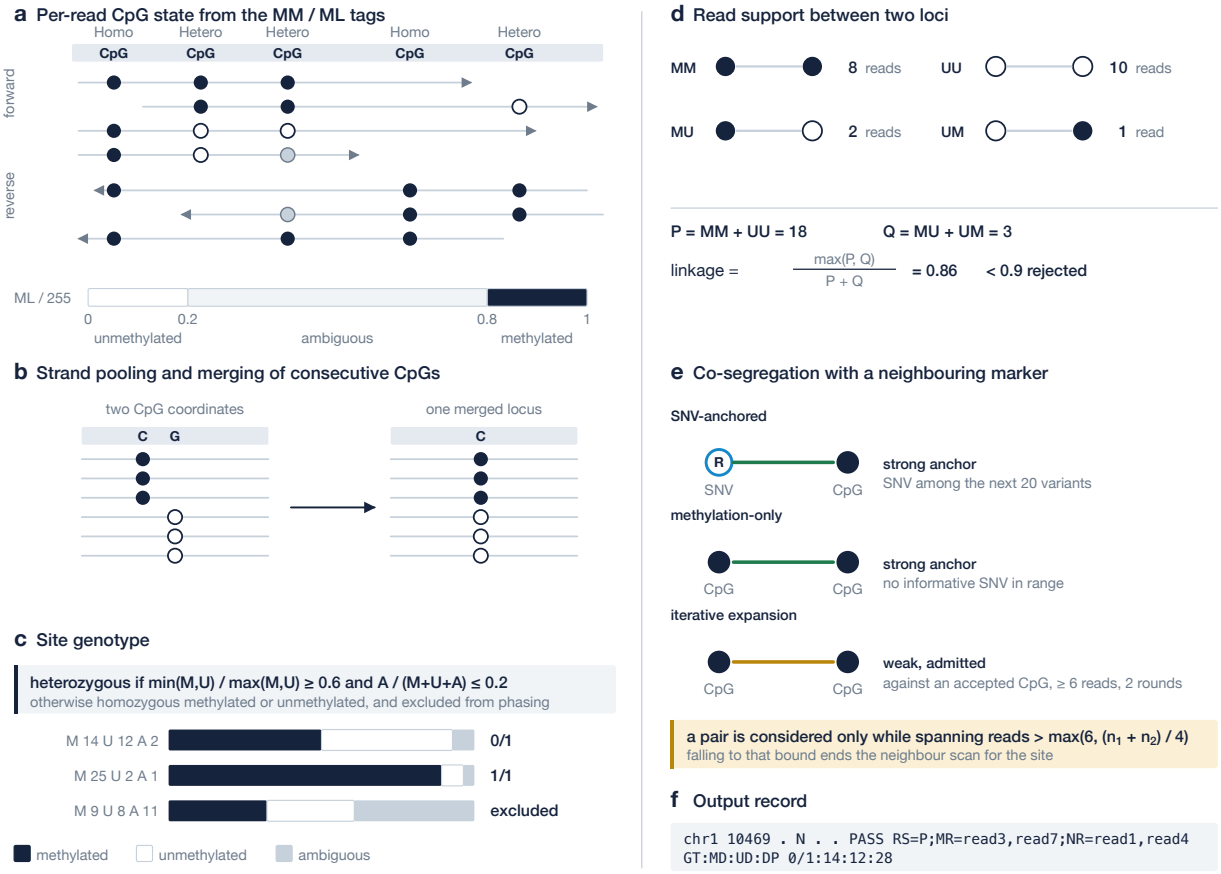
Supplementary Fig. 4: Overview of the graph neural network correction module. **a**, Every phased variant with a phasing entropy of at least 0.8 (amber) triggers a window of ±20 variants. **b**, The window is represented as a graph of allele nodes with 31 node features and six edge features (Supplementary Tables 1 and 2); squares represent features, not their values. **c**, Each of four GPS layers combines a local GATv2 branch with a global self-attention branch; a classifier head assigns each allele node a score for a label-defined phase-disagreement onset. **d**, Variants with an onset score of at least 0.35 are unphased without changing their genotypes, and the phase set is split where the remaining graph is disconnected. Details are shown in Supplementary Figs. 5 and 6.



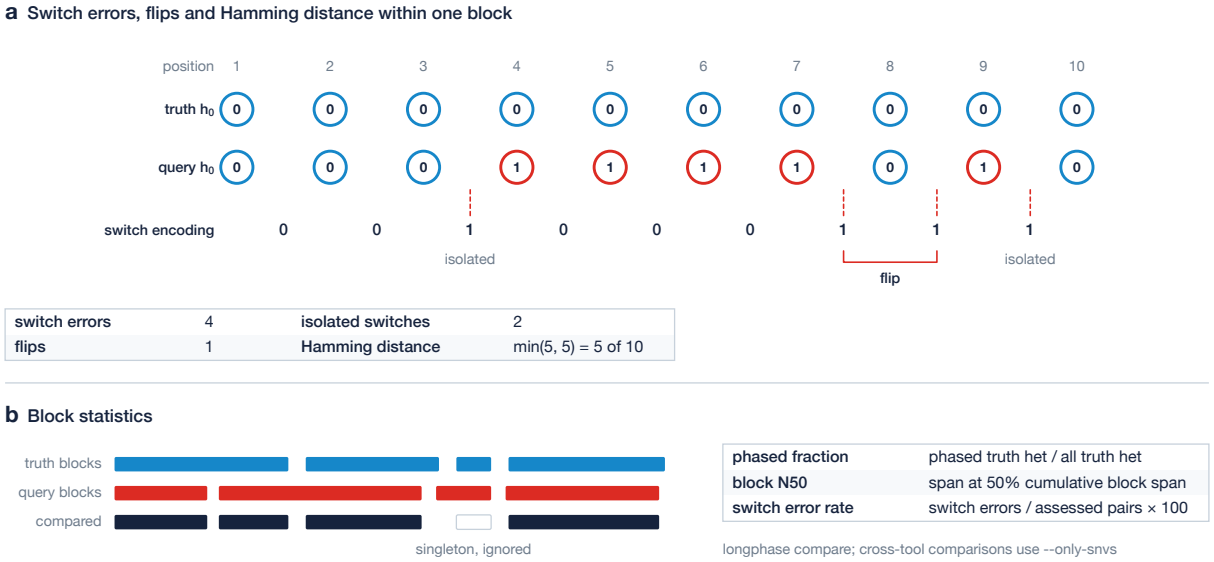
Supplementary Fig. 5: Window construction and phase-set update for the graph neural network. **a**, Window of ±20 variants around a trigger with PE ≥ 0.8; arcs are phasing-graph edges and the dashed grey edge is a cross-haplotype link. **b**, Bridge-vertex feature: a variant whose removal disconnects the window graph. **c**, Only variants within half the largest distance from the trigger contribute to the training loss and to inference scores; the other variants supply graph context. **d**, The phasing graph in its DOT export format (`-dot`), with the vote weight at the source as the edge label. **e**, Before correction: a variant with an onset score ≥ 0.35 (amber) inside a phase set. **f**, After correction: the variant is unphased; the largest remaining component keeps the original PS and every other component receives the position of its first variant as its PS.



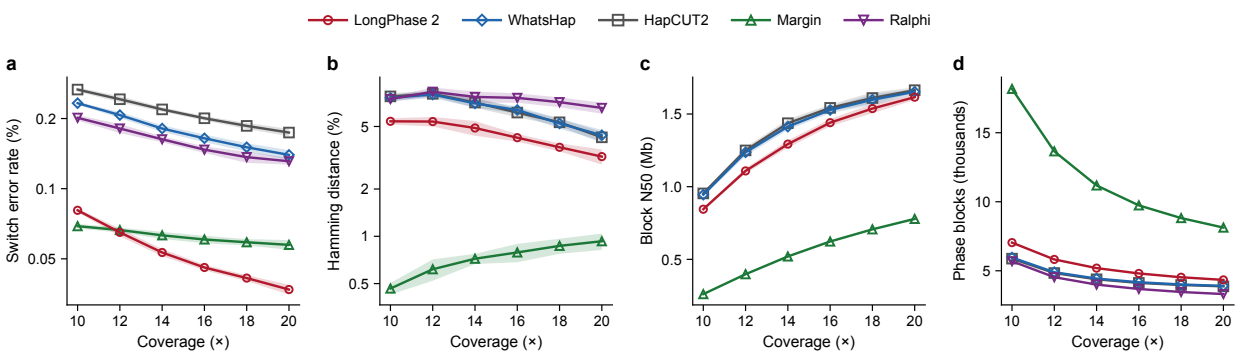
Supplementary Fig. 6: Graph neural network architecture. Node features are standardized and projected to 128 dimensions. Each of four GPS layers combines a local GATv2 branch over the window graph, with edge features entering through a per-layer edge encoder, and a global self-attention branch over all nodes; the branches are added to the residual stream and followed by layer normalization and a feed-forward block. The final embedding is concatenated with the sum and the maximum of the logistic-transformed weights $\sigma(w) = 1/(1 + e^{-w})$ of the incoming edges, excluding the self-loop, and passed to a two-class classifier whose error-class output is the onset score; the scores of the two allele nodes of a variant are averaged.



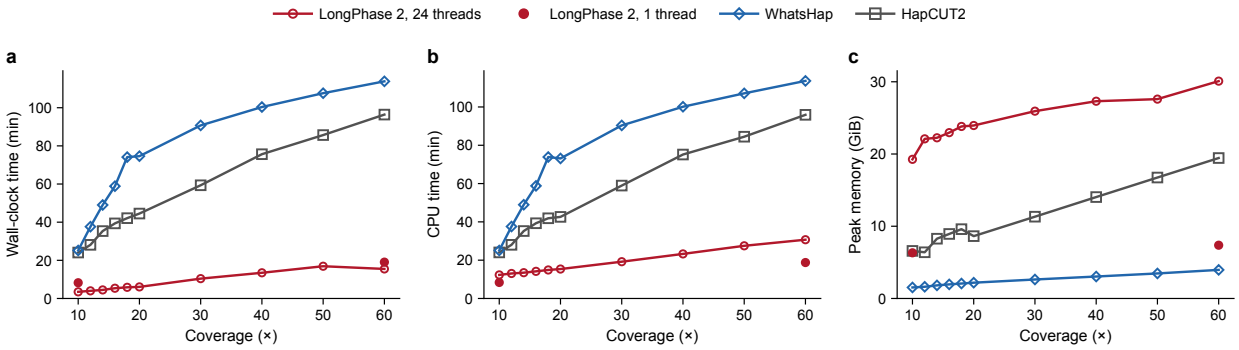
Supplementary Fig. 7: Allele-specific methylation calling with `modcall`. **a**, Per-read CpG state from the `MM`/`ML` tags: methylated at `ML`/255 ≥ 0.8, unmethylated at ≤ 0.2, ambiguous in between. **b**, Strand-specific CpG observations are combined into loci for phasing: reverse-strand calls are mapped to the forward CpG coordinate and consecutive CpGs are merged. **c**, A CpG is genotyped as a candidate allele-specific site (0/1) when $\min(M,U)/\max(M,U) \geq 0.6$ and $A/(M+U+A) \leq 0.2$, for methylated, unmethylated and ambiguous read counts M, U and A; other sites are excluded. **d**, Read support between two loci in the four state pairs; a pair is linked when $\max(P,Q)/(P+Q) \geq 0.9$, and the example shown is rejected. **e**, Linking against an SNV among the next 20 variants, against another candidate CpG where no SNV is in range, and expansion to weak candidates (Supplementary Method 1). **f**, Output record; the `MR` and `NR` read names attach the 5mC alleles to reads.



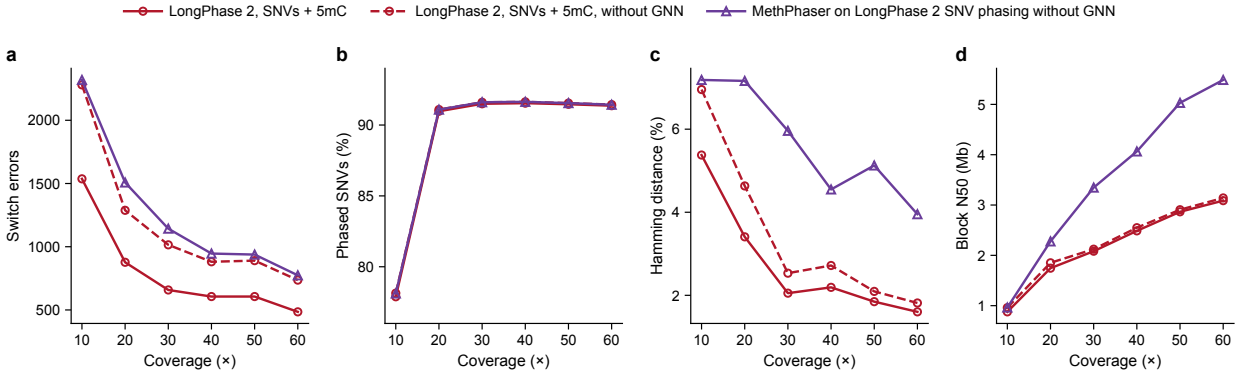
Supplementary Fig. 8: Phasing evaluation metrics. **a**, Within one intersected block, the query haplotype string is compared with the truth in switch encoding; the number of differing positions is the switch-error count, adjacent switch pairs are counted as flips, and the Hamming distance is the minimum over the two orientations of misassigned variants. **b**, Block intersection and the block-level statistics. Only variants that are heterozygous in both files and phased in both are compared (dark bars); blocks left with a single variant are ignored (outline). Switch error rate and Hamming distance are computed over the intersected blocks; the phased fraction counts truth heterozygous variants matched and phased in the query before the partition, and the block N50 is computed on the query's own blocks of at least two variants, as implemented in `longphase compare`.



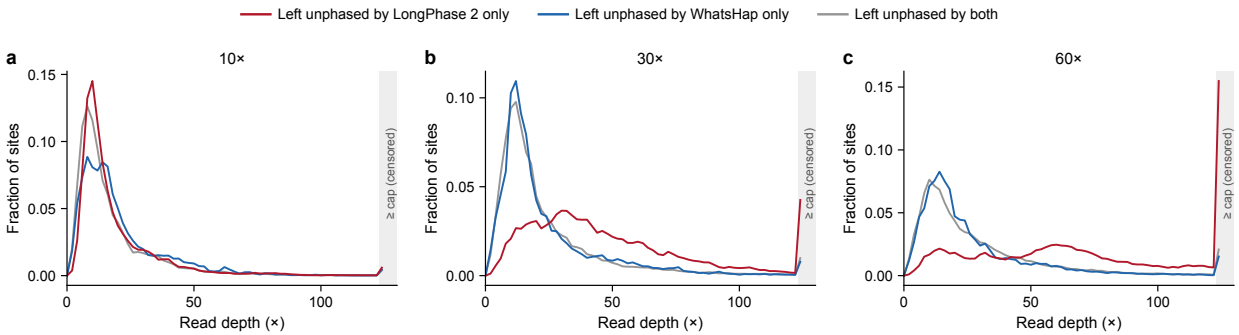
Supplementary Fig. 9: SNV-only phasing compared with Margin and Ralphi. HG002 nanopore R10.4.1 data at 10–20×, means of ten replicates with s.d., scored against v5.0q; LongPhase 2 with GNN correction. **a**, Switch error rate. **b**, Hamming distance. **c**, Block N50; Ralphi is omitted because its blocks sum to more than the autosome length. **d**, Number of phase blocks. Values in Supplementary Table 8.



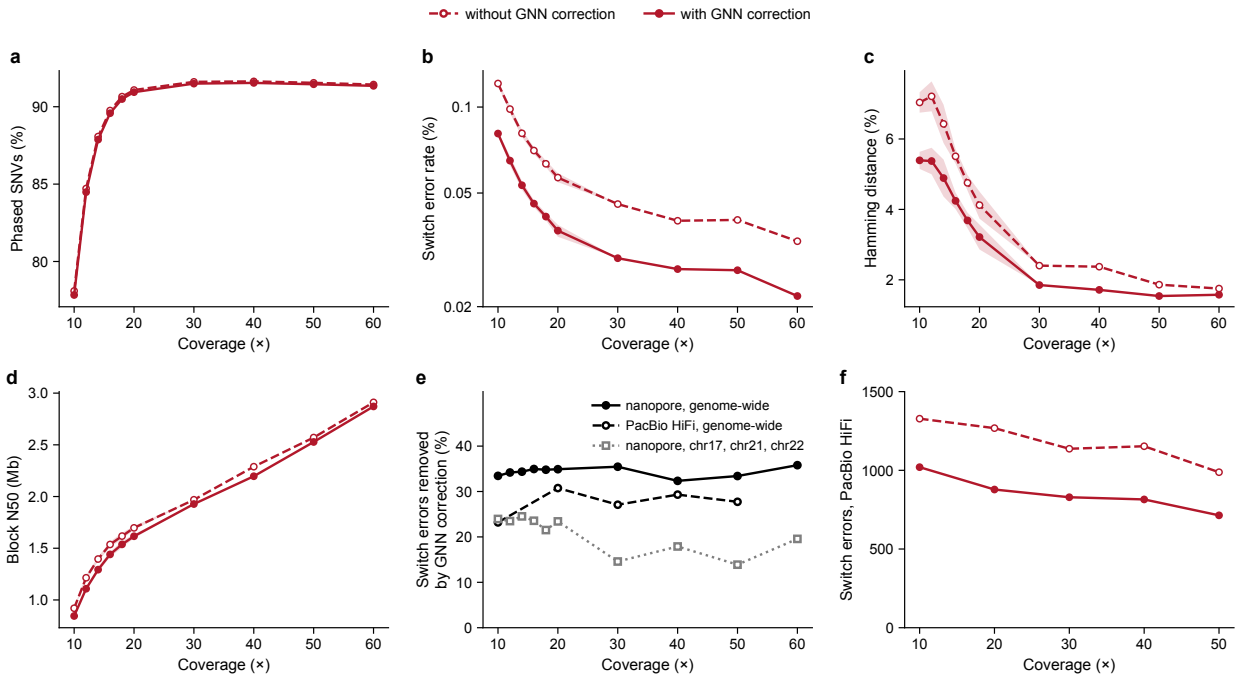
Supplementary Fig. 10: Runtime and peak memory of SNV-only phasing. HG002 nanopore R10.4.1 data, replicate 1, on the workstation described in Methods; values in Supplementary Table 16. LongPhase 2 includes GNN correction and was run with 24 threads at every coverage and on one thread at 10 and 60×; WhatsHap and HapCUT2 run on one thread. HapCUT2 at 30 and 40× was re-timed (Methods). **a**, Wall-clock time. **b**, CPU time (user plus system). **c**, Peak resident memory.



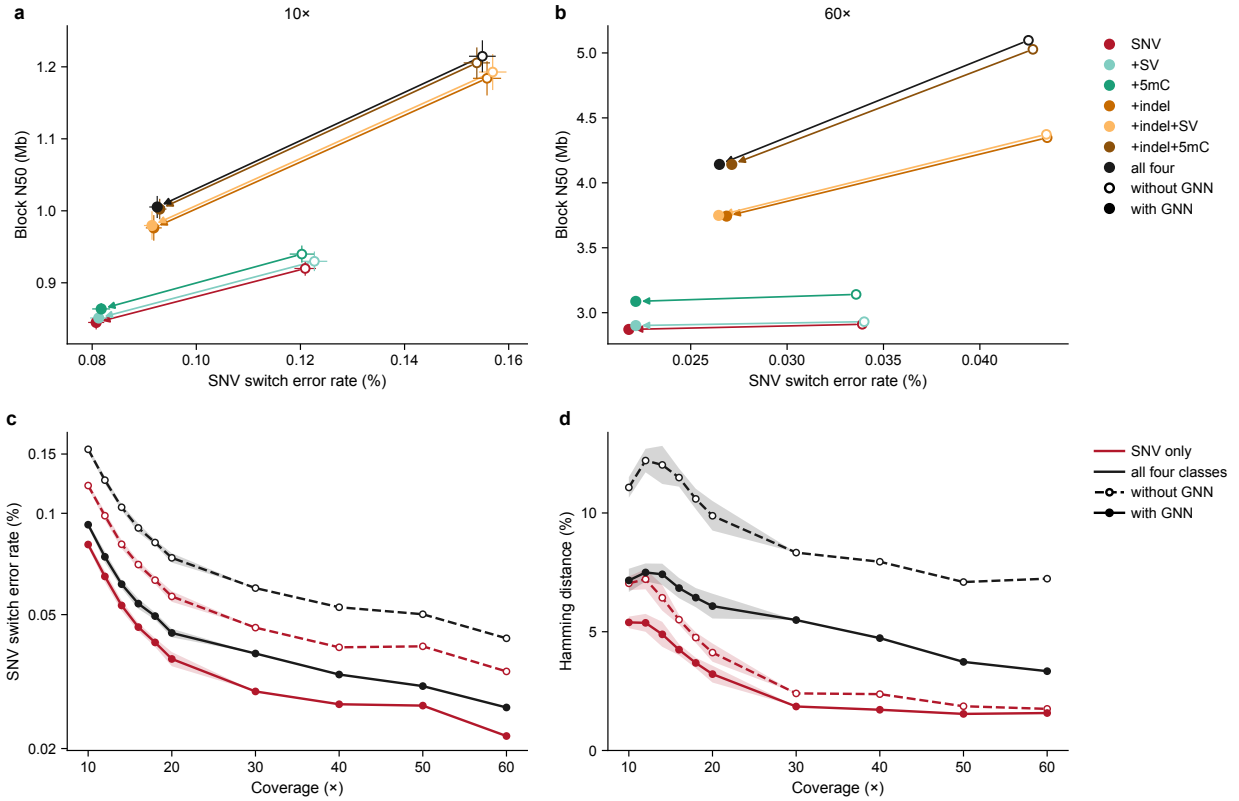
Supplementary Fig. 11: Joint methylation co-phasing compared with MethPhaser. HG002 nanopore R10.4.1 data, replicate 1, scored on SNVs against v5.0q. MethPhaser was given the SNV-only output of LongPhase 2 without GNN correction. **a**, Switch errors. **b**, Fraction of benchmark heterozygous SNVs phased. **c**, Hamming distance. **d**, Block N50.



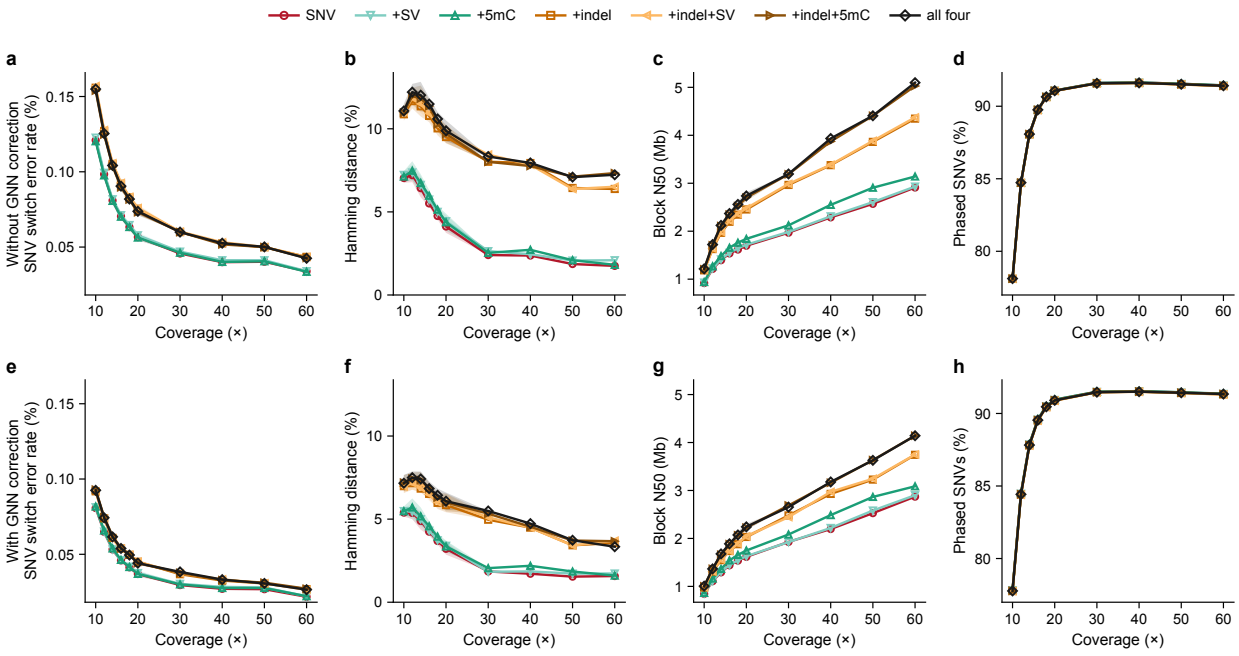
Supplementary Fig. 12: Read depth at heterozygous SNV calls left unphased by LongPhase 2 only, by WhatsHap only or by both. HG002 nanopore R10.4.1 data, replicate 1, at 10× (**a**), 30× (**b**) and 60× (**c**); 2× bins. The grey bin collects every site at or above the depth cap of the variant caller (about 125×), whose true depth is censored. Sets as in Supplementary Table 17.



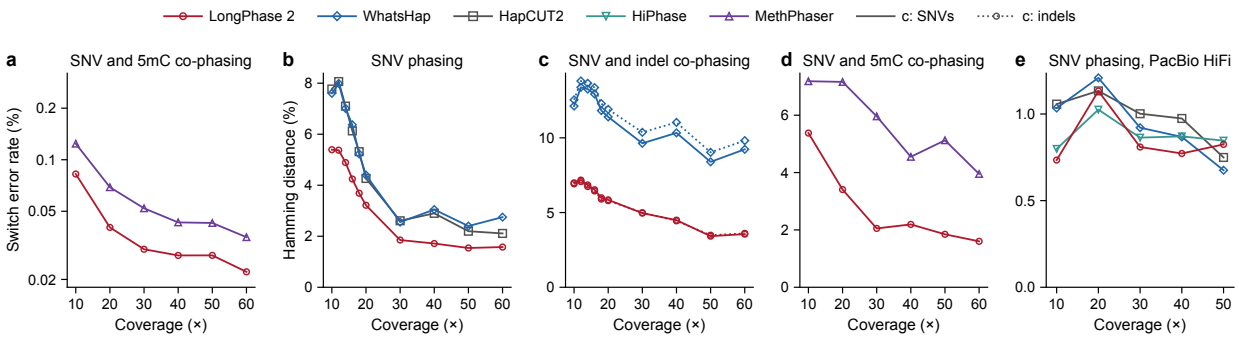
Supplementary Fig. 13: Effect of GNN correction on SNV-only phasing. LongPhase 2 with and without GNN correction; HG002 nanopore R10.4.1 data unless stated; means of ten replicates with s.d. at 10–20×; one down-sampling at 30–60×. **a**, Fraction of benchmark heterozygous SNVs phased. **b**, Switch error rate. **c**, Hamming distance. **d**, Block N50. **e**, Fraction of switch errors removed by GNN correction on nanopore and PacBio HiFi data, genome-wide, and on the nanopore chromosomes held out from GNN training (chr17, chr21 and chr22; Supplementary Table 15). **f**, Switch errors on PacBio HiFi data, with the network trained on nanopore data.



Supplementary Fig. 14: Co-phasing configurations with and without GNN correction. HG002 nanopore R10.4.1 data; seven combinations of variant classes, each phased separately; SNV metrics against v5.0q. **a**,**b**, SNV switch error rate against block N50 at 10× (**a**; means of ten replicates with s.d.) and 60× (**b**); arrows run from the result without (open) to the result with (filled) GNN correction. **c**,**d**, SNV switch error rate (**c**) and Hamming distance (**d**; computed over SNVs and indels for four-class phasing) for SNV-only and four-class phasing.



Supplementary Fig. 15: All co-phasing configurations across coverage. HG002 nanopore R10.4.1 data; means of ten replicates with s.d. at 10–20×; one down-sampling at 30–60×. Top row without, bottom row with GNN correction. **a**,**e**, SNV switch error rate. **b**,**f**, Hamming distance (over SNVs and indels where indels are phased). **c**,**g**, Block N50. **d**,**h**, Fraction of benchmark heterozygous SNVs phased. Values in Supplementary Table 10.



Supplementary Fig. 16: Switch error rate of SNV and 5mC co-phasing and block-wise Hamming distance of the phaser comparisons in Fig. 2, across coverage. Samples, coverages and replicates as in Fig. 2; scored against v5.0q. **a**, Switch error rate of SNV and 5mC co-phasing compared with MethPhaser, replicate 1 (log scale; block N50 in Fig. 2d). **b**–**e**, Hamming distance. **b**, SNV phasing of nanopore data compared with WhatsHap and HapCUT2. **c**, SNV and indel co-phasing compared with WhatsHap; solid lines score SNVs and dotted lines indels (Supplementary Table 14). **d**, SNV and 5mC co-phasing compared with MethPhaser, replicate 1. **e**, SNV phasing of PacBio HiFi data compared with WhatsHap, HapCUT2 and HiPhase.

References

[1] Rampášek, L. *et al.* Recipe for a general, powerful, scalable graph transformer. In *Advances in Neural Information Processing Systems*, vol. 35 (2022).

[2] Veličković, P. *et al.* Graph attention networks. In *International Conference on Learning Representations (ICLR)* (2018).

[3] Brody, S., Alon, U. & Yahav, E. How attentive are graph attention networks? In *International Conference on Learning Representations (ICLR)* (2022).

[4] Vaswani, A. *et al.* Attention is all you need. In *Advances in Neural Information Processing Systems*, vol. 30 (2017).

[5] Lin, T.-Y., Goyal, P., Girshick, R., He, K. & Dollár, P. Focal loss for dense object detection. In *IEEE International Conference on Computer Vision (ICCV)* (2017).

[6] Patterson, M. *et al.* WhatsHap: weighted haplotype assembly for future-generation sequencing reads. *Journal of Computational Biology* **22**, 498–509 (2015).

[7] Shafin, K. *et al.* Haplotype-aware variant calling with PEPPER-Margin-DeepVariant enables high accuracy in nanopore long-reads. *Nature Methods* **18**, 1322–1332 (2021).

[8] Hansen, N. F. *et al.* A complete diploid human genome benchmark for personalized genomics. *Cell* **189**, 4857–4875.e31 (2026).

[9] Olson, N. D. *et al.* The Genome in a Bottle HG002 assembly-based variant benchmark set enables comprehensive benchmarking of small and structural variants. *bioRxiv* (2026).

[10] Chen, H., Nie, F., Luo, F. & Wang, J. HapBridge: a methylation-guided approach for correcting switch errors and bridging phased blocks in long-read phasing. *bioRxiv* (2025).

[11] Pfennig, A. & Akey, J. M. Methylation-aware long-read phasing significantly improves genome-wide haplotype reconstruction. *bioRxiv* (2026).

[12] Hartmann, U., Shaham, E., Nathan, D., Blech, I. & Zeevi, D. TinkerHap—a novel read-based phasing algorithm with integrated multimethod support for enhanced accuracy. *GigaScience* **14**, giaf138 (2025).

[13] Holt, J. M. *et al.* HiPhase: jointly phasing small, structural, and tandem repeat variants from HiFi sequencing. *Bioinformatics* **40** (2024).

[14] Zheng, Z. *et al.* ClairS: a deep-learning method for long-read tumor–normal pair somatic small variant calling. *Nature Methods* **23**, 1507–1520 (2026).

[15] Chen, L. *et al.* ClairS-TO: a deep-learning method for long-read tumor-only somatic small variant calling. *Nature Communications* **16** (2025).

[16] Zheng, Z. *et al.* Clair3-RNA: a deep learning-based small variant caller for long-read RNA sequencing data. *Nature Communications* **16** (2025).

[17] Zheng, Z. *et al.* Accelerated long-read variant calling with Clair3 for whole-genome sequencing. *Bioinformatics* **42** (2026).

[18] Keskus, A. G. *et al.* Severus detects somatic structural variation and complex rearrangements in cancer genomes using long-read sequencing. *Nature Biotechnology* **44**, 247–257 (2025).

[19] Oxford Nanopore Technologies. epi2me-labs/wf-human-variation: Nextflow workflow for the analysis of human genetic variation from Oxford Nanopore sequencing data. `https://github.com/epi2me-labs/wf-human-variation` (2025). Version 2.x; phasing performed with LongPhase (`-use_longphase`).

[20] Pacific Biosciences. HiFi-somatic-WDL: tumor-normal variant calling workflow using HiFi reads. `https://github.com/PacificBiosciences/HiFi-Somatic-WDL` (2025).

[21] Cao, S. *et al.* cuteHap: haplotype-aware structural variant detection in phased long-read sequencing data. *Advanced Science* **13** (2026).

[22] Wang, S. *et al.* A scalable framework for comprehensive typing of polymorphic immune genes from long-read data. *Advanced Science* **13** (2026).

[23] Gabbutt, C. *et al.* Fluctuating DNA methylation tracks cancer evolution at clinical scale. *Nature* **645**, 764–773 (2025).

[24] O'Neill, K. *et al.* Long-read sequencing of an advanced cancer cohort resolves rearrangements, unravels haplotypes, and reveals methylation landscapes. *Cell Genomics* **4**, 100674 (2024).

[25] LaFlamme, C. W. *et al.* Diagnostic utility of DNA methylation analysis in genetically unsolved pediatric epilepsies and CHD2 episignature refinement. *Nature Communications* **15** (2024).

[26] Iwasaki, Y. *et al.* Bidirectional disruption of *GNAS* transcripts causes broad methylation defects in pseudohypoparathyroidism type 1B. *Proceedings of the National Academy of Sciences* **122** (2025).

[27] Zalusky, M. P. *et al.* 3-hour genome sequencing and targeted analysis to rapidly assess genetic risk. *Genetics in Medicine Open* **2**, 101833 (2024).

[28] Smits, D. J. *et al.* Nanopore long-read sequencing for the critically ill facilitates ultrarapid diagnostics and urgent clinical decision making. *European Journal of Human Genetics* **34**, 108–118 (2025).

[29] Fleming, A. *et al.* Combined approaches, including long-read sequencing, address the diagnostic challenge of HYDIN in primary ciliary dyskinesia. *European Journal of Human Genetics* **32**, 1074–1085 (2024).

[30] Thomas, R. *et al.* Splice-modulating antisense oligonucleotides targeting a pathogenic intronic variant in adult polyglucosan body disease correct mis-splicing and restore enzyme activity in patient cells. *Nucleic Acids Research* **53** (2025).

[31] Tian, Y., McDonnell, S. K., Wu, L., Larson, N. B. & Wang, L. Fine mapping regulatory variants by characterizing native CpG methylation with nanopore long-read sequencing. *Human Genetics and Genomics Advances* **7**, 100532 (2026).

[32] Galvez, J. H. *et al.* Benchmarking of Oxford Nanopore whole genome sequencing for germline variant and CpG methylation detection across Canada's national platform for genome sequencing and analysis. *Frontiers in Genetics* **17** (2026).

[33] Flack, N. *et al.* Chromosome-level, nanopore-only genome and allele-specific DNA methylation of Pallas's cat, *Otocolobus manul*. *NAR Genomics and Bioinformatics* **5** (2023).

[34] Battistella, E., Maheshwari, A., Ekim, B., Berger, B. & Popic, V. Graph-based deep reinforcement learning for haplotype assembly with Ralphi. *Genome Research* **35**, 2617–2625 (2025).

[35] Luo, J., Wang, J., Zhai, H. & Wang, J. GCphase: an SNP phasing method using a graph partition and error correction algorithm. *BMC Bioinformatics* **25** (2024).

[36] Zhou, Y., Leung, A. W.-S., Ahmed, S. S., Lam, T.-W. & Luo, R. Duet: SNP-assisted structural variant calling and phasing using Oxford nanopore sequencing. *BMC Bioinformatics* **23** (2022).